

1. Questions

Read the following information carefully and answer the questions based on it.

There are five boats A, B, C, D, and E travel a certain distance in five different rivers on a particular day. The speed of the river in which boat B travels is $\frac{X}{100}$ % more than the speed of the river in which boat A travels. The speed of boat A in still water is 20 km/hr, which is $\frac{2}{3}$ of the speed of boat B in still water. The ratio of the upstream speed of boat A and the downstream speed of boat B is 5:12. The downstream distance covered by boat B in 4 hours is 54 km more than the upstream distance covered by boat A in 6 hours. The speed of the river in which boat C travels is 33.33% more than the speed of the river in which boat B travels. The downstream speed of boat C is $\frac{Y}{100}$ km/hr. Boat C can cover 48 km upstream distance in 3 hours. The downstream speed of boat D is 50% of the downstream speed of boat C. The time taken by boat D to cover 120 km in still water is twice that taken by boat C to cover the same distance in still water. The speed of boat E in still water is 25% more than that of boat D. The speed of the river in which boat E travels is $\frac{3}{4}$ of the speed of the river in which boat D travels. The upstream distance covered by boat D in 9 hours is equal to the downstream distance covered by boat E in $\frac{Z}{100}$ hours.

Note:

The speed of the river in which each boat travels may not be the same.

The speed of each boat in still water is also different.

If the speed of the river in which boat A is travelling is increased by $\frac{V}{100}$ km/hr, then the ratio of the time taken by boat A to travel 140 km downstream and 120 km upstream will be 1:2. Find the percentage change in the upstream speed of boat A.

- a. 20% increase
- b. 20% decrease
- c. 50% increase
- d. 50% decrease
- e. None of these

2. Questions

If boat B travels T hours downstream and $T+3$ hours upstream, then the downstream distance covered by boat B will be 12 km less than the upstream distance covered by the same boat. Find the time taken by boat B to cover $18T$ km in still water.

- a. 5 hours
- b. 8 hours
- c. 2 hours
- d. 3 hours

@QuizByMickey

e. None of these

3. Questions

The speed of boat F in still water is 10 km/hr more than boat E, while the speed of the river in which boat F is traveling is [P] km/hr. If the boat F covers [3Q] km downstream in 2 hours and [Q] km upstream in 1 hour, then find the value of [Q/P].

- a. 4
- b. 5
- c. 12
- d. 18
- e. None of these

4. Questions

If the speed of boat D in still water is reduced by 3 km/hr and the speed of the river in which boat D is traveling is also reduced by 25%, then which of the following statements (s) is/are definitely true?

Statement I: The time taken by boat D to cover 60 km upstream distance is 10 hours.

Statement II: The ratio of the downstream and upstream speed of boat D will be 2:1.

Statement III: If boat D travels for 3 hours in still water, then the total distance covered will be 25 km.

- a. Only statement I
- b. Both statements II and III
- c. Both statements I and II
- d. All statements I, II, and III
- e. None of the statements is true

5. Questions

In an another river, if boat E covers 4 hours downstream and 7 hours upstream, then the total distance covered by boat E downstream will be 23 km less than the total distance covered by boat E upstream. Find the time taken by boat E to cover 85 km downstream in another river.

- a. 10 hours
- b. 5 hours
- c. 15 hours
- d. 8 hours
- e. 7 hours

6. Questions

Read the following information carefully and answer the questions based on it.

The table chart given below shows the percentage distribution of the number of people tested for COVID in each zone out of the total number of people tested for COVID in five different zones together. It also shows the ratio of the number of people tested positive and negative, and the ratio of the number of people who are vaccinated and those who are not vaccinated in each zone.

Zone	The percentage of the total number of people tested	The ratio of the number of people who tested positive and negative	The ratio of the number of people who are vaccinated and those who are not vaccinated
North		11:7	6:5
East	24%		
Central		2:1	
West		5:4	
South	13%		11:3

Notes:

- i). Some data in the table chart is missing.
- ii). The total number of people tested = the number of people who tested positive + the number of people who tested negative
- iii). The total number of people who tested positive = the number of people who are vaccinated + the number of people who are not vaccinated
- iv). In the north zone, the difference between the number of people who tested negative and those who are not vaccinated is 24.
- v). The average number of people who tested positive in the central and west zones is 160.
- vi). The number of people who tested negative in the west zone is 100 more than the number of people who tested negative in the central zone.

In the east zone, if the number of people who tested negative is twice the number of people who are not vaccinated, which is 48 less than the number of people who are vaccinated, then find the total number of people who tested positive in the east zone.

- a. 144
- b. 168
- c. 120
- d. 108
- e. 128

7. Questions

If the number of people who tested negative in south zone is $[X]$ less than west zone, while the number of people who tested positive in south zone is $[X-76]$ more than the number of people who tested negative in the same zone, then find the number of people who are vaccinated in south zone.

- a. $(X/2)$
- b. $(X+11)$
- c. $(X-22)$
- d. Cannot be determined
- e. $(X + 2)$

8. Questions

In the central zone, the ratio of the number of males and females who are vaccinated and those who are not vaccinated is 5:3 and 1:1, respectively, while the number of males who are vaccinated is 21 more than the number of females who are not vaccinated, then find the number of people who are vaccinated in the central zone.

- a. 72
- b. 36
- c. 48
- d. 60
- e. 81

9. Questions

In the north zone, out of the total number of people(died + alive) who are not vaccinated, if _____% of people die, then the difference between the number of people who died and those who are alive will be _____.

Which of the following satisfies the filler(s) in the same order?

I: 10 and 40

II: 25 and 30

III: 30 and 24

- a. Only II follow
- b. Both I and III follow
- c. Both II and III follow
- d. All I, II, and III follow
- e. None follow

@QuizByMickey

10. Questions

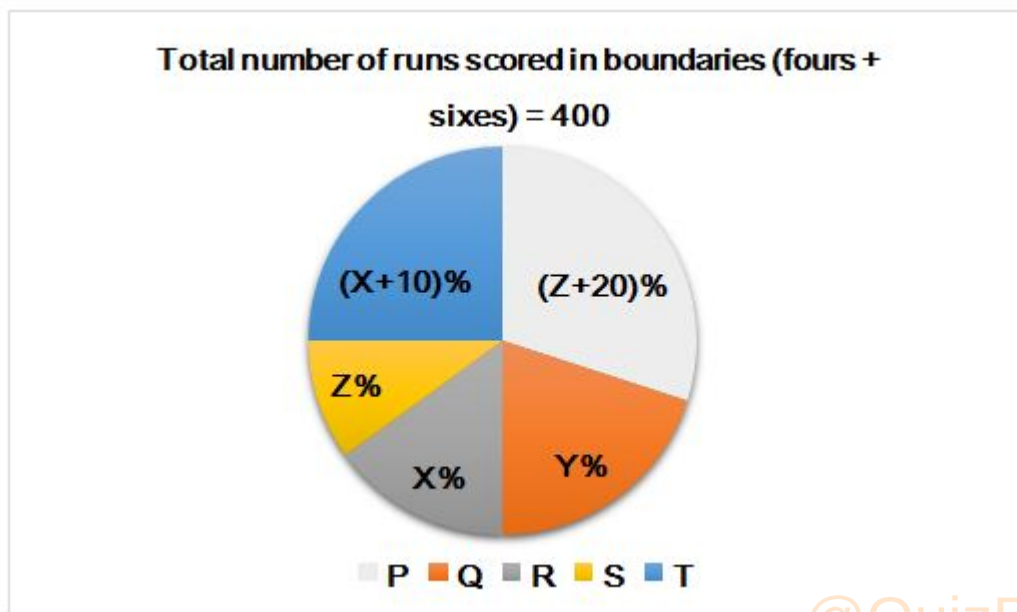
Which of the following is/are definitely true?

- I:** The number of people who tested negative in the north zone is 40% more than central zone.
- II:** The ratio of the total number of people who tested in the central and the west zone is 2:1, respectively.
- III:** The number of people who tested positive in the west zone is 225.
- Only I
 - Both I and II
 - Both II and III
 - All I, II, and III
 - None is true

11. Questions

Read the following information carefully and answer the questions based on it.

The pie chart given below shows the percentage distribution of the total number of runs scored in boundaries (fours + sixes) by five batters P, Q, R, S, and T.



Notes:

The total number of runs scored by any batter = the number of runs scored in boundaries + the number of runs scored other than boundaries

The total number of runs scored in boundaries = the total number of runs scored in fours + the total number of runs scored in sixes

The strike-rate = $\left[\left\{ \frac{\text{the total number of runs scored}}{\text{the total number of balls faced}} \right\} \times 100 \right]$

A batter will get 4 runs for each four and 6 runs for each six scored by him.

The total number of runs scored in boundaries (fours + sixes) by all five batters together is 400.

The ratio of the number of fours and sixes hit by P, Q, R, S, and T is 7:2, 1:1, 6:1, 7:2, and 19:4, respectively.

The total number of runs scored by P in fours is 36 more than the total number of runs scored by Q in sixes

The total number of runs scored by P in sixes is 4 more than the total number of runs scored by Q in fours.

If the number of runs scored other than boundaries by P is [K] and the total number of balls faced by P is 240, then his strike-rate will be [K-160]. Find the number of runs scored in boundaries by P is what percent of the total number of runs scored (boundaries + other than boundaries) by him?

- a. 25%
- b. 40%
- c. 15%
- d. 50%
- e. 75%

12. Questions

If the strike-rate of Q is 80, while the number of runs scored other than boundaries by Q is _____% of the total number of runs scored (boundaries + other than boundaries) by him, then the total number of balls faced by Q will be _____.

Which of the following satisfies the filler(s) in the same order?

I: 80 and 500

II: 75 and 400

III: 60 and 250

- a. Both I and II
- b. Only I
- c. Only III
- d. All I, II, and III
- e. Only II

@QuizByMickey

13. Questions

Out of the total number of runs scored other than boundaries by R, the ratio of the number of singles, doubles, and triples is 5:10:1. If the total number of balls faced by R is 160 and his strike-rate is 125, then find the number of dot balls faced by R.

- a. 66
- b. 72

- c. 80
- d. 54
- e. 84

14. Questions

If the total number of runs scored (boundaries + other than boundaries) by S and T together is [L], while $\frac{3}{5}$ th and 80% of the total number of runs scored (boundaries + other than boundaries) by S and T respectively are scored other than boundaries, then [L] will be exactly divisible by ____[?].

What should come in place of the question mark?

- a. 18
- b. 12
- c. 16
- d. Both a) and b)
- e. Both b) and c)

15. Questions

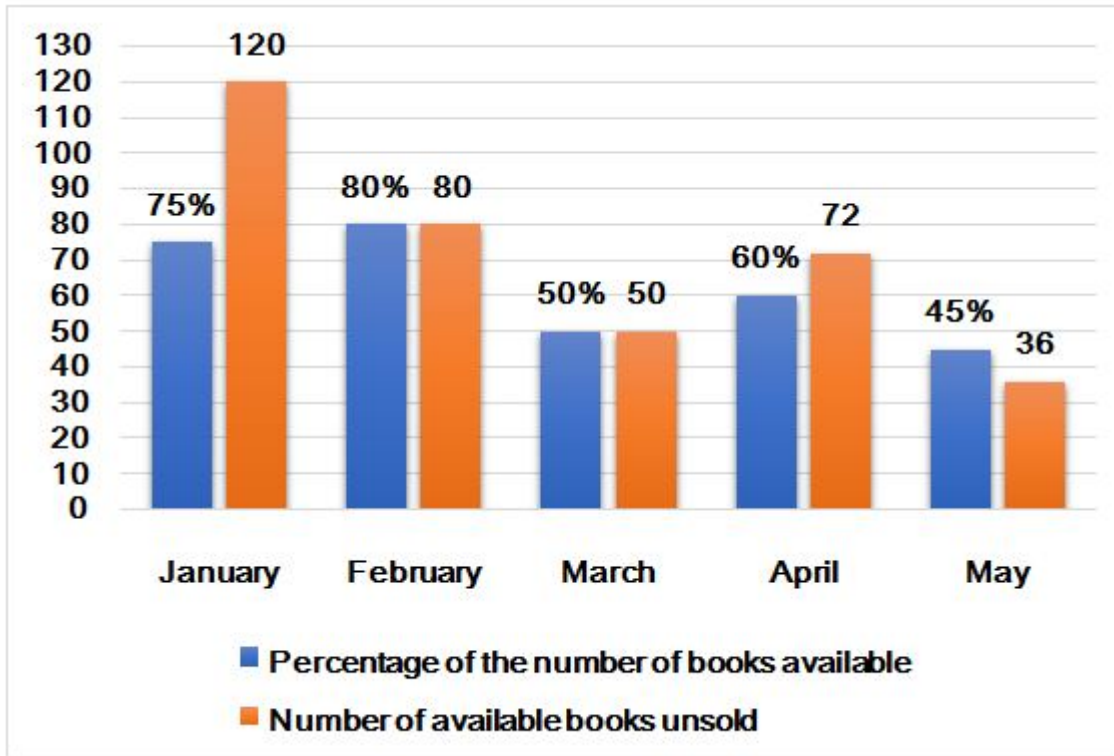
The total number of runs scored by another batter U is [28X], while the ratio of the number of runs scored in boundaries to the runs scored other than boundaries is [F:G]. If the number of fours and sixes scored by U is 23 and 8, respectively, then determine the correct relation between [F] and [G].

- a. $F > G$
- b. $F < G$
- c. $F \geq G$
- d. $F \leq G$
- e. $F = G$

16. Questions

Read the following information carefully and answer the questions based on it.

The bar graph given below shows the percentage of the number of **available books** out of the total number of books published (available + unavailable) by a bookstore in five different months, January, February, March, April, and May of the year 2024. It also shows the **number of available books that remained unsold** in each month.



Note:

The **total number of books published** = the number of **available books** + the number of **unavailable books**.

The total number of **available books**= the number of books sold + the number of books unsold

Only available books can be sold.

The ratio of the total number of books published in January to February is 4:5.

The difference between the number of available books sold in January and February is 140. (January < February)

The total number of books published in April is 50% more than the previous month.

The number of available books sold in March is 74 is less than in April.

In May, the ratio of the number of available books sold to the number of **unavailable books** is 5:11.

___[A]___ % of the total number of books published in January are non-fictional. If the number of non-fictional books published in February is 200, which is ___[B]___ more than the number of non-fictional books published in January, then the ratio of the number of fictional books published in January to February will be 14:15. (Only fictional and non-fictional books are published in January and February)

Which of the following equation(s) is/are correct?

I: $(A+50) = B$

II: $(B/2) = (A+10)$

III: $(A+B) = 120$

- a. Only II follow
- b. Both I and II follow
- c. Only III follow
- d. All I, II, and III follow
- e. None follow

17. Questions

In March, if the price of each available book is Rs. 50, which is sold at a profit of $[X]\%$, then there will be an overall selling price of all sold books is Rs. 5600. Now, if $[X]$ is divided by 3 and then 50 is added, then the resultant will be $[Y]$.

Find the value of $[\sqrt{(5X-3Y+31)}]$.

- a. 11
- b. 15
- c. 20
- d. 24
- e. None of these

18. Questions

In another month, June, the total number of books published is 40 less than in March, while the number of unavailable books in June is $[Z]$, which is 24 more than the number of available books unsold in the same month. If the ratio of the number of available books sold to unsold in June is 9:1. Find Z

- a. 60
- b. 30
- c. 75
- d. 25
- e. 40

@QuizByMickey

19. Questions

If the number of poetry books available in May is 30% of the previous month, and the number of story books available in April is 72 more than in May, then find the total number of story books sold in April and May together. (Only story and poetry books are available in April and May)

- a. 160
- b. 120
- c. 240

- d. 200
- e. None of these

20. Questions

In July, the number of available books sold to males is 40 more than those sold to females, which is equal to the number of available books unsold. If the total number of available books (sold + unsold) in June is 100, then find the number of available books sold to males in June.

- a. 60
- b. 42
- c. 36
- d. 24
- e. 70

Explanations:

1. Questions

The speed of boat A in still water = 20 km/hr

The speed of boat B in still water = $20 * (3/2) = 30$ km/hr

Now, let the upstream speed of boat A and the downstream speed of boat B be $5v$ km/hr and $12v$ km/hr, respectively.

So,

The downstream distance covered by boat B in 4 hours = $(12v \times 4) = 48v$ km/hr

The upstream distance covered by boat A in 6 hours = $(5v \times 6) = 30v$ km/hr

Now,

$$48v - 30v = 54$$

$$18v = 54$$

$$v = 3$$

Boat A

The speed of the boat in still water = 20 km/hr

The upstream speed = $5v = (5 \times 3) = 15$ km/hr

The speed of the river = $(20 - 15) = 5$ km/hr

The downstream speed = $(20 + 5) = 25$ km/hr

Boat B

@QuizByMickey

The speed of the boat in still water = 30 km/hr

The downstream speed = $12v = (12 \times 3) = 36$ km/hr

The speed of the river = $(36-30) = 6$ km/hr

The upstream speed = $(30-6) = 24$ km/hr

Now, the percentage by which the speed of the river in which boat B travels is more than the speed of the river in which boat A travels = $\left[\frac{(6-5)}{5} \times 100\right]\% = \left\{\frac{(1)}{5} \times 100\right\}\% = 20\%$

So, $X = 20$

Boat C

The speed of the river in which boat C travel = $\frac{4}{3}$ of 6 = 8 km/hr (**Note: $33.33\% = \frac{1}{3}$**)

The upstream speed of boat C = $(48/3) = 16$ km/hr

So, the speed of boat C in still water = $(16 + 8) = 24$ km/hr

Therefore, the downstream speed of boat C = $(24 + 8) = 32$ km/hr

So, $Y = 32$

Boat D

The downstream speed of boat D = 50% of 32 = 16 km/hr

The time taken by boat C to cover 120 km in still water = $(120/24) = 5$ hours

So, the time taken by boat D to cover 120 km in still water = $(5 \times 2) = 10$ hours

Now, the speed of boat D in still water = $(120/10) = 12$ km/hr

Downstream of boat D (Boat+River) = 50% of downstream of C = $\frac{1}{2} \times 32 = 16$ km/hr

So, the speed of the river in which boat D travels = $(16-12) = 4$ km/hr

And, the upstream speed of boat D = $(12-4) = 8$ km/hr

Boat E

The speed of boat E in still water = 125% of 12 = 15 km/hr

The speed of the river in which boat E travels = $\frac{3}{4}$ of 4 = 3 km/hr

The downstream speed of boat E = $(15+3) = 18$ km/hr

The upstream speed of boat E = $(15-3) = 12$ km/hr

The upstream distance covered by boat D in 9 hours = $(8 \times 9) = 72$ km

So, the downstream distance covered by boat E = 72 km

Therefore, the time taken by boat E to cover the downstream distance = $(72/18) = 4$ hours

So, $Z = 4$

Now, the values of,

@QuizByMickey

$$X = 20$$

$$Y = 32$$

$$Z = 4$$

Boat	The speed of the boat in still water (in km/hr)	The speed of the river (in km/hr)	The downstream speed (in km/hr)	The upstream speed (in km/hr)
A	20	5	25	15
B	30	6	36	24
C	24	8	32	16
D	12	4	16	8
E	15	3	18	12

Answer: B

According to the question,

Boat A

The speed of the boat in still water = 20 km/hr

The speed of the river = (5+V) km/hr

The downstream speed = {20+(5+V)} = (25+V) km/hr

The upstream speed = {20-(5+V)} = (15-V) km/hr

Now,

$$\{140/(25+V)\}/\{120/(15-V)\} = 1/2$$

$$280/(25+V) = 120/(15-V)$$

$$4200 - 280V = 3000 + 120V$$

$$400V = 1200$$

$$V = 3$$

Now, the initial upstream speed of boat A = 15 km/hr

And, the new upstream speed of boat A = (15-V) = (15-3) = 12 km/hr

Therefore, the percentage change = $\left[\frac{(15-12)}{15} \times 100\right]\% = \left\{\frac{(3)}{15} \times 100\right\}\% = 20\%$ decrease.

Hence, the correct answer is option B.

2. Questions

The speed of boat A in still water = 20 km/hr

The speed of boat B in still water = $20 \times \frac{3}{2} = 30$ km/hr

Now, let the upstream speed of boat A and the downstream speed of boat B be 5v km/hr and 12v km/hr,

respectively.

So,

The downstream distance covered by boat B in 4 hours = $(12v \times 4) = 48v$ km/hr

The upstream distance covered by boat A in 6 hours = $(5v \times 6) = 30v$ km/hr

Now,

$$48v - 30v = 54$$

$$18v = 54$$

$$v = 3$$

Boat A

The speed of the boat in still water = 20 km/hr

The upstream speed = $5v = (5 \times 3) = 15$ km/hr

The speed of the river = $(20 - 15) = 5$ km/hr

The downstream speed = $(20 + 5) = 25$ km/hr

Boat B

The speed of the boat in still water = 30 km/hr

The downstream speed = $12v = (12 \times 3) = 36$ km/hr

The speed of the river = $(36 - 30) = 6$ km/hr

The upstream speed = $(30 - 6) = 24$ km/hr

Now, the percentage by which the speed of the river in which boat B travels is more than the speed of the river in which boat A travels = $\left[\frac{(6-5)}{5} \times 100\right]\% = \left\{\frac{(1)}{5} \times 100\right\}\% = 20\%$

So, $X = 20$

Boat C

The speed of the river in which boat C travel = $\frac{4}{3}$ of 6 = 8 km/hr (*Note: 33.33% = 1/3*)

The upstream speed of boat C = $(48/3) = 16$ km/hr

So, the speed of boat C in still water = $(16 + 8) = 24$ km/hr

@QuizByMickey

Therefore, the downstream speed of boat C = $(24 + 8) = 32$ km/hr

So, $Y = 32$

Boat D

The downstream speed of boat D = 50% of 32 = 16 km/hr

The time taken by boat C to cover 120 km in still water = $(120/24) = 5$ hours

So, the time taken by boat D to cover 120 km in still water = $(5 \times 2) = 10$ hours

Now, the speed of boat D in still water = $(120/10) = 12$ km/hr

Downstream of boat D (Boat+River) = 50% of downstream of C = $1/2 * 32 = 16$ km/hr

So, the speed of the river in which boat D travels = $(16-12) = 4$ km/hr

And, the upstream speed of boat D = $(12-4) = 8$ km/hr

Boat E

The speed of boat E in still water = 125% of 12 = 15 km/hr

The speed of the river in which boat E travels = $3/4$ of 4 = 3 km/hr

The downstream speed of boat E = $(15+3) = 18$ km/hr

The upstream speed of boat E = $(15-3) = 12$ km/hr

The upstream distance covered by boat D in 9 hours = $(8 \times 9) = 72$ km

So, the downstream distance covered by boat E = 72 km

Therefore, the time taken by boat E to cover the downstream distance = $(72/18) = 4$ hours

So, $Z = 4$

Now, the values of,

$X = 20$

$Y = 32$

$Z = 4$

Boat	The speed of the boat in still water (in km/hr)	The speed of the river (in km/hr)	The downstream speed (in km/hr)	The upstream speed (in km/hr)
A	20	5	25	15
B	30	6	36	24
C	24	8	32	16
D	12	4	16	8
E	15	3	18	12

Answer: D

@QuizByMickey

Boat B

The downstream speed = 36 km/hr

The upstream speed = 24 km/hr

Now,

$$\{(T+3) * 24\} - (T * 36) = 12$$

$$(24T+72) - 36T = 12$$

$$24T + 60 = 36T$$

$$12T = 60$$

$$T = 5$$

Now, the speed of boat B in still water = 30 km/hr

Therefore, the time taken by boat B to cover [18T] km in still water = $\{(18 * 5)/30\} = (90/30) = 3$ hours.

Hence, the correct answer is option D.

3. Questions

The speed of boat A in still water = 20 km/hr

The speed of boat B in still water = $20 * (3/2) = 30$ km/hr

Now, let the upstream speed of boat A and the downstream speed of boat B be $5v$ km/hr and $12v$ km/hr, respectively.

So,

The downstream distance covered by boat B in 4 hours = $(12v \times 4) = 48v$ km/hr

The upstream distance covered by boat A in 6 hours = $(5v \times 6) = 30v$ km/hr

Now,

$$48v - 30v = 54$$

$$18v = 54$$

$$v = 3$$

Boat A

The speed of the boat in still water = 20 km/hr

The upstream speed = $5v = (5 * 3) = 15$ km/hr

The speed of the river = $(20 - 15) = 5$ km/hr

The downstream speed = $(20 + 5) = 25$ km/hr

Boat B

The speed of the boat in still water = 30 km/hr

The downstream speed = $12v = (12 * 3) = 36$ km/hr

The speed of the river = $(36 - 30) = 6$ km/hr

The upstream speed = $(30 - 6) = 24$ km/hr

Now, the percentage by which the speed of the river in which boat B travels is more than the speed of the river in which boat A travels = $[\{(6 - 5)/5\} \times 100]\% = \{(1/5) \times 100\}\% = 20\%$

So, X = 20

Boat C

@QuizByMickey

The speed of the river in which boat C travel = $\frac{4}{3}$ of 6 = 8 km/hr (*Note: 33.33% = $\frac{1}{3}$*)

The upstream speed of boat C = $(48/3) = 16$ km/hr

So, the speed of boat C in still water = $(16 + 8) = 24$ km/hr

Therefore, the downstream speed of boat C = $(24 + 8) = 32$ km/hr

So, $Y = 32$

Boat D

The downstream speed of boat D = 50% of 32 = 16 km/hr

The time taken by boat C to cover 120 km in still water = $(120/24) = 5$ hours

So, the time taken by boat D to cover 120 km in still water = $(5 \times 2) = 10$ hours

Now, the speed of boat D in still water = $(120/10) = 12$ km/hr

Downstream of boat D (Boat+River) = 50% of downstream of C = $\frac{1}{2} \times 32 = 16$ km/hr

So, the speed of the river in which boat D travels = $(16 - 12) = 4$ km/hr

And, the upstream speed of boat D = $(12 - 4) = 8$ km/hr

Boat E

The speed of boat E in still water = 125% of 12 = 15 km/hr

The speed of the river in which boat E travels = $\frac{3}{4}$ of 4 = 3 km/hr

The downstream speed of boat E = $(15 + 3) = 18$ km/hr

The upstream speed of boat E = $(15 - 3) = 12$ km/hr

The upstream distance covered by boat D in 9 hours = $(8 \times 9) = 72$ km

So, the downstream distance covered by boat E = 72 km

Therefore, the time taken by boat E to cover the downstream distance = $(72/18) = 4$ hours

So, $Z = 4$

Now, the values of,

$X = 20$

$Y = 32$

$Z = 4$

@QuizByMickey

Boat	The speed of the boat in still water (in km/hr)	The speed of the river (in km/hr)	The downstream speed (in km/hr)	The upstream speed (in km/hr)
A	20	5	25	15
B	30	6	36	24
C	24	8	32	16
D	12	4	16	8
E	15	3	18	12

Answer: A

The speed of boat E in still water = 15 km/hr

Boat F

The speed of the boat in still water = $(15 + 10) = 25$ km/hr

The speed of the river = P km/hr

The downstream speed = $(25+P)$ km/hr

The upstream speed = $(25-P)$ km/hr

Now,

$$3Q/(25+P) = 2$$

$$50 + 2P = 3Q$$

$$3Q - 2P = 50 \dots\dots\dots(i)$$

And,

$$Q/(25-P) = 1$$

$$25 - P = Q$$

$$Q + P = 25 \dots\dots\dots(ii)$$

By solving equations (i) and (ii), we get,

$$3Q - 2P = 50$$

$$3Q + 3P = 75$$

$$5P = 25$$

$$P = 5$$

Now, put the value of P in equation (i), we get,

$$3Q - 2P = 50$$

$$3Q - (2 \times 5) = 50$$

$$3Q = 50 + 10$$

@QuizByMickey

$$3Q = 60$$

$$Q = 20$$

Therefore, the value of $(Q/P) = (20/5) = 4$.

Hence, the correct answer is option A.

4. Questions

The speed of boat A in still water = 20 km/hr

The speed of boat B in still water = $20 * (3/2) = 30$ km/hr

Now, let the upstream speed of boat A and the downstream speed of boat B be $5v$ km/hr and $12v$ km/hr, respectively.

So,

The downstream distance covered by boat B in 4 hours = $(12v \times 4) = 48v$ km/hr

The upstream distance covered by boat A in 6 hours = $(5v \times 6) = 30v$ km/hr

Now,

$$48v - 30v = 54$$

$$18v = 54$$

$$v = 3$$

Boat A

The speed of the boat in still water = 20 km/hr

The upstream speed = $5v = (5 \times 3) = 15$ km/hr

The speed of the river = $(20 - 15) = 5$ km/hr

The downstream speed = $(20 + 5) = 25$ km/hr

Boat B

The speed of the boat in still water = 30 km/hr

The downstream speed = $12v = (12 \times 3) = 36$ km/hr

The speed of the river = $(36 - 30) = 6$ km/hr

The upstream speed = $(30 - 6) = 24$ km/hr

Now, the percentage by which the speed of the river in which boat B travels is more than the speed of the river in which boat A travels = $[(6 - 5)/5] \times 100\% = \{(1/5) \times 100\}\% = 20\%$

So, $X = 20$

Boat C

The speed of the river in which boat C travel = $4/3$ of $6 = 8$ km/hr (**Note: $33.33\% = 1/3$**)

The upstream speed of boat C = $(48/3) = 16$ km/hr

@QuizByMickey

So, the speed of boat C in still water = $(16 + 8) = 24$ km/hr

Therefore, the downstream speed of boat C = $(24 + 8) = 32$ km/hr

So, $Y = 32$

Boat D

The downstream speed of boat D = 50% of 32 = 16 km/hr

The time taken by boat C to cover 120 km in still water = $(120/24) = 5$ hours

So, the time taken by boat D to cover 120 km in still water = $(5 \times 2) = 10$ hours

Now, the speed of boat D in still water = $(120/10) = 12$ km/hr

Downstream of boat D (Boat+River) = 50% of downstream of C = $1/2 \times 32 = 16$ km/hr

So, the speed of the river in which boat D travels = $(16-12) = 4$ km/hr

And, the upstream speed of boat D = $(12-4) = 8$ km/hr

Boat E

The speed of boat E in still water = 125% of 12 = 15 km/hr

The speed of the river in which boat E travels = $3/4$ of 4 = 3 km/hr

The downstream speed of boat E = $(15+3) = 18$ km/hr

The upstream speed of boat E = $(15-3) = 12$ km/hr

The upstream distance covered by boat D in 9 hours = $(8 \times 9) = 72$ km

So, the downstream distance covered by boat E = 72 km

Therefore, the time taken by boat E to cover the downstream distance = $(72/18) = 4$ hours

So, $Z = 4$

Now, the values of,

$X = 20$

$Y = 32$

$Z = 4$

@QuizByMickey

Boat	The speed of the boat in still water (in km/hr)	The speed of the river (in km/hr)	The downstream speed (in km/hr)	The upstream speed (in km/hr)
A	20	5	25	15
B	30	6	36	24
C	24	8	32	16
D	12	4	16	8
E	15	3	18	12

Answer: C

Boat D

The reduced speed of boat D in still water = $(12-3) = 9$ km/hr

The reduced speed of the river = 75% of 4 = 3 km/hr

The downstream speed = $(9+3) = 12$ km/hr

The upstream speed = $(9-3) = 6$ km/hr

Statement I: The time taken by boat D to cover the upstream distance = $(60/6) = 10$ hours.

So, statement I is definitely true.

Statement II: The ratio of the downstream and upstream distance = $12:6 = 2:1$

So, statement II is definitely true.

Statement III: The distance covered in still water = $(3 \times 9) = 27$ km

So, statement III is definitely not true.

Therefore, both statements I and II are definitely true.

Hence, the correct answer is option C.

5. Questions

The speed of boat A in still water = 20 km/hr

The speed of boat B in still water = $20 \times (3/2) = 30$ km/hr

Now, let the upstream speed of boat A and the downstream speed of boat B be $5v$ km/hr and $12v$ km/hr, respectively.

So,

The downstream distance covered by boat B in 4 hours = $(12v \times 4) = 48v$ km/hr

The upstream distance covered by boat A in 6 hours = $(5v \times 6) = 30v$ km/hr

Now,

$$48v - 30v = 54$$

$$18v = 54$$

$$v = 3$$

Boat A

The speed of the boat in still water = 20 km/hr

The upstream speed = $5v = (5 \times 3) = 15$ km/hr

The speed of the river = $(20-15) = 5$ km/hr

The downstream speed = $(20+5) = 25$ km/hr

@QuizByMickey

Boat B

The speed of the boat in still water = 30 km/hr

The downstream speed = $12v = (12 \times 3) = 36$ km/hr

The speed of the river = $(36 - 30) = 6$ km/hr

The upstream speed = $(30 - 6) = 24$ km/hr

Now, the percentage by which the speed of the river in which boat B travels is more than the speed of the river in which boat A travels = $\left[\frac{(6-5)}{5} \times 100\right]\% = \left\{\frac{(1)}{5} \times 100\right\}\% = 20\%$

So, $X = 20$

Boat C

The speed of the river in which boat C travel = $\frac{4}{3}$ of 6 = 8 km/hr (**Note: $33.33\% = \frac{1}{3}$**)

The upstream speed of boat C = $(48/3) = 16$ km/hr

So, the speed of boat C in still water = $(16 + 8) = 24$ km/hr

Therefore, the downstream speed of boat C = $(24 + 8) = 32$ km/hr

So, $Y = 32$

Boat D

The downstream speed of boat D = 50% of 32 = 16 km/hr

The time taken by boat C to cover 120 km in still water = $(120/24) = 5$ hours

So, the time taken by boat D to cover 120 km in still water = $(5 \times 2) = 10$ hours

Now, the speed of boat D in still water = $(120/10) = 12$ km/hr

Downstream of boat D (Boat+River) = 50% of downstream of C = $\frac{1}{2} \times 32 = 16$ km/hr

So, the speed of the river in which boat D travels = $(16 - 12) = 4$ km/hr

And, the upstream speed of boat D = $(12 - 4) = 8$ km/hr

Boat E

The speed of boat E in still water = 125% of 12 = 15 km/hr

The speed of the river in which boat E travels = $\frac{3}{4}$ of 4 = 3 km/hr

The downstream speed of boat E = $(15 + 3) = 18$ km/hr

The upstream speed of boat E = $(15 - 3) = 12$ km/hr

The upstream distance covered by boat D in 9 hours = $(8 \times 9) = 72$ km

So, the downstream distance covered by boat E = 72 km

Therefore, the time taken by boat E to cover the downstream distance = $(72/18) = 4$ hours

So, $Z = 4$

@QuizByMickey

Now, the values of,

$$X = 20$$

$$Y = 32$$

$$Z = 4$$

Boat	The speed of the boat in still water (in km/hr)	The speed of the river (in km/hr)	The downstream speed (in km/hr)	The upstream speed (in km/hr)
A	20	5	25	15
B	30	6	36	24
C	24	8	32	16
D	12	4	16	8
E	15	3	18	12

Answer: B

Boat D

Let the speed of the river be R km/hr

The speed of boat E in still water = 15 km/hr

The downstream speed = $(15+R)$ km/hr

The upstream speed = $(15-R)$ km/hr

Now,

$$\{(15-R) * 7\} - \{(15+R) * 4\} = 23$$

$$(105-7R) - (60+4R) = 23$$

$$82 - 7R = 60 + 4R$$

$$11R = 22$$

$$R = 2$$

Now, the downstream speed of boat E = $(15+R) = (15+2) = 17$ km/hr

Therefore, the time taken by boat E to cover the downstream distance = $(85/17) = 5$ hours.

Hence, the correct answer is option B.

6. Questions

Let the number of people who tested positive and negative in the north zone be 11m and 7m, respectively.

So, the number of people who are vaccinated in the north zone = $6/11$ of $11m = 6m$

And, the number of people who are not vaccinated in the north zone = $5/11$ of $11m = 5m$

Now,

$$7m - 5m = 24$$

$$2m = 24$$

$$m = 12$$

North Zone

The number of people who tested positive = $11m = (11 * 12) = 132$

The number of people who tested negative = $7m = (7 * 12) = 84$

So, the total number of people who tested = $(132+84) = 216$

The number of people who are vaccinated = $6m = (6 * 12) = 72$

The number of people who are not vaccinated = $5m = (5 * 12) = 60$

Now, the ratio of the number of people who tested positive and negative in the central zone = 2:1. Let it be 2a and a, respectively.

And, the ratio of the number of people who tested positive and negative in the west zone = 5:4. Let it be 5b and 4b, respectively.

So,

$$5b + 2a = (160 * 2)$$

$$5b + 2a = 320 \dots \dots \dots (I)$$

Also,

$$4b - a = 100 \dots \dots \dots (II)$$

By solving equations (I) and (II), we get,

$$20b + 8a = 1280$$

$$20b - 5a = 500$$

$$13a = 780$$

$$a = 60$$

Now, put the value of a in equation (I), we get,

$$5b + 2a = 320$$

$$5b + (2 * 60) = 320$$

$$5b = 320 - 120$$

$$5b = 200$$

$$b = 40$$

Central Zone

The number of people who tested positive = $2a = (2 * 60) = 120$

The number of people who tested negative = $a = 60$

@QuizByMickey

So, the total number of people who tested = $(120+60) = 180$

West Zone

The number of people who tested positive = $5b = (5 * 40) = 200$

The number of people who tested negative = $4b = (4 * 40) = 160$

So, the total number of people who tested = $(200+160) = 360$

Now,

The total percentage distribution of the total number of people who tested (positive + negative) from all five zones together = 100%

And, the sum of the percentage distribution of the total number of people who tested (positive + negative) from the north, central, and west zone = $\{100-(24+13)\}\% = (100-37)\% = 63\%$

The total number of people who tested (positive + negative) from north, central and west zone = $(216+180+360) = 756$

Now,

$$63\% = 756$$

$$1\% = 12$$

$$100\% = (12 \times 100) = 1200$$

Therefore, the total number of people who tested (positive + negative) from all five zones together = 1200

Now, the total number of people who tested (positive + negative) from east zone = 24% of 1200 = 288

And, the total number of people who tested (positive + negative) from south zone = 13% of 1200 = 156

Answer: B

According to the question,

East Zone

The total number of people who tested = 288

Let the number of people who are not vaccinated be n .

So, the number of people who tested negative = $(2 * n) = 2n$

The number of people who tested positive = $(288 - 2n)$

Now,

$$(288 - 2n) - 48 = n$$

$$288 - 48 = 4n$$

$$4n = 240$$

$$n = 60$$

Therefore, the number of people who tested positive = $(288 - 2n) = [288 - (2 * 60)] = (288 - 120) = 168$.

@QuizByMickey

Hence, the correct answer is option B.

7. Questions

Let the number of people who tested positive and negative in the north zone be $11m$ and $7m$, respectively.

So, the number of people who are vaccinated in the north zone = $6/11$ of $11m = 6m$

And, the number of people who are not vaccinated in the north zone = $5/11$ of $11m = 5m$

Now,

$$7m - 5m = 24$$

$$2m = 24$$

$$m = 12$$

North Zone

The number of people who tested positive = $11m = (11 * 12) = 132$

The number of people who tested negative = $7m = (7 * 12) = 84$

So, the total number of people who tested = $(132+84) = 216$

The number of people who are vaccinated = $6m = (6 * 12) = 72$

The number of people who are not vaccinated = $5m = (5 * 12) = 60$

Now, the ratio of the number of people who tested positive and negative in the central zone = $2:1$. Let it be $2a$ and a , respectively.

And, the ratio of the number of people who tested positive and negative in the west zone = $5:4$. Let it be $5b$ and $4b$, respectively.

So,

$$5b + 2a = (160 * 2)$$

$$5b + 2a = 320 \dots \dots \dots (I)$$

Also,

$$4b - a = 100 \dots \dots \dots (II)$$

By solving equations (I) and (II), we get,

$$20b + 8a = 1280$$

$$20b - 5a = 500$$

$$13a = 780$$

$$a = 60$$

Now, put the value of a in equation (I), we get,

$$5b + 2a = 320$$

$$5b + (2 * 60) = 320$$

@QuizByMickey

$$5b = 320 - 120$$

$$5b = 200$$

$$b = 40$$

Central Zone

$$\text{The number of people who tested positive} = 2a = (2 * 60) = 120$$

$$\text{The number of people who tested negative} = a = 60$$

$$\text{So, the total number of people who tested} = (120+60) = 180$$

West Zone

$$\text{The number of people who tested positive} = 5b = (5 * 40) = 200$$

$$\text{The number of people who tested negative} = 4b = (4 * 40) = 160$$

$$\text{So, the total number of people who tested} = (200+160) = 360$$

Now,

The total percentage distribution of the total number of people who tested (positive + negative) from all five zones together = 100%

And, the sum of the percentage distribution of the total number of people who tested (positive + negative) from the north, central, and west zone = $\{100 - (24+13)\}\% = (100-37)\% = 63\%$

$$\text{The total number of people who tested (positive + negative) from north, central and west zone} = (216+180+360) = 756$$

Now,

$$63\% = 756$$

$$1\% = 12$$

$$100\% = (12 \times 100) = 1200$$

Therefore, the total number of people who tested (positive + negative) from all five zones together = 1200

Now, the total number of people who tested (positive + negative) from east zone = 24% of 1200 = 288

And, the total number of people who tested (positive + negative) from south zone = 13% of 1200 = 156

Answer: C

The number of people who tested negative in West Zone = 160

South Zone

$$\text{The number of people who tested negative} = (160-X)$$

$$\text{The number of people who tested positive} = \{(160-X) + (X-76)\} = 84$$

$$\text{The total number of people who tested} = 156$$

Now,

$$(160-X) + 84 = 156$$

$$160 - X = 72$$

$$X = 88$$

So, the number of people who are vaccinated in the south zone = $11/14$ of $84 = 66$.

$$(X-22) = (88-22) = 66$$

Hence, the correct answer is option C.

8. Questions

Let the number of people who tested positive and negative in the north zone be $11m$ and $7m$, respectively.

So, the number of people who are vaccinated in the north zone = $6/11$ of $11m = 6m$

And, the number of people who are not vaccinated in the north zone = $5/11$ of $11m = 5m$

Now,

$$7m - 5m = 24$$

$$2m = 24$$

$$m = 12$$

North Zone

The number of people who tested positive = $11m = (11 * 12) = 132$

The number of people who tested negative = $7m = (7 * 12) = 84$

So, the total number of people who tested = $(132+84) = 216$

The number of people who are vaccinated = $6m = (6 * 12) = 72$

The number of people who are not vaccinated = $5m = (5 * 12) = 60$

Now, the ratio of the number of people who tested positive and negative in the central zone = $2:1$. Let it be $2a$ and a , respectively.

And, the ratio of the number of people who tested positive and negative in the west zone = $5:4$. Let it be $5b$ and $4b$, respectively.

So,

$$5b + 2a = (160 * 2)$$

$$5b + 2a = 320 \dots \dots \dots (I)$$

Also,

$$4b - a = 100 \dots \dots \dots (II)$$

By solving equations (I) and (II), we get,

$$20b + 8a = 1280$$

$$20b - 5a = 500$$

@QuizByMickey

$$13a = 780$$

$$a = 60$$

Now, put the value of a in equation (I), we get,

$$5b + 2a = 320$$

$$5b + (2 * 60) = 320$$

$$5b = 320 - 120$$

$$5b = 200$$

$$b = 40$$

Central Zone

The number of people who tested positive = $2a = (2 * 60) = 120$

The number of people who tested negative = $a = 60$

So, the total number of people who tested = $(120+60) = 180$

West Zone

The number of people who tested positive = $5b = (5 * 40) = 200$

The number of people who tested negative = $4b = (4 * 40) = 160$

So, the total number of people who tested = $(200+160) = 360$

Now,

The total percentage distribution of the total number of people who tested (positive + negative) from all five zones together = 100%

And, the sum of the percentage distribution of the total number of people who tested (positive + negative) from the north, central, and west zone = $\{100 - (24+13)\} \% = (100-37)\% = 63\%$

The total number of people who tested (positive + negative) from north, central and west zone = $(216+180+360) = 756$

Now,

$$63\% = 756$$

$$1\% = 12$$

$$100\% = (12 \times 100) = 1200$$

Therefore, the total number of people who tested (positive + negative) from all five zones together = 1200

Now, the total number of people who tested (positive + negative) from east zone = 24% of 1200 = 288

And, the total number of people who tested (positive + negative) from south zone = 13% of 1200 = 156

Answer: A

Central Zone

@QuizByMickey

Let the number of males and females who are vaccinated be $5p$ and $3p$, respectively.

Also, let the number of males and females who are not vaccinated be q and q , respectively.

Now,

$$5p - q = 21 \dots\dots\dots(i)$$

And,

$$5p + 3p + q + q = 120$$

$$8p + 2q = 120$$

$$4p + q = 60 \dots\dots\dots(ii)$$

By solving equations (i) and (ii), we get,

$$9p = 81$$

$$p = 9$$

and

$$36 + q = 60$$

$$q = 24$$

Therefore,

$$\text{The number of males who are vaccinated} = 5p = 45$$

$$\text{The number of females who are vaccinated} = 3p = 27$$

$$\text{The number of males who are not vaccinated} = q = 24$$

$$\text{The number of females who are not vaccinated} = q = 24$$

$$\text{Therefore, the number of people who are vaccinated} = 45 + 27 = 72$$

Hence, the correct answer is option A.

9. Questions

Let the number of people who tested positive and negative in the north zone be $11m$ and $7m$, respectively.

$$\text{So, the number of people who are vaccinated in the north zone} = \frac{6}{11} \text{ of } 11m = 6m$$

$$\text{And, the number of people who are not vaccinated in the north zone} = \frac{5}{11} \text{ of } 11m = 5m$$

Now,

$$7m - 5m = 24$$

$$2m = 24$$

$$m = 12$$

North Zone

$$\text{The number of people who tested positive} = 11m = (11 * 12) = 132$$

The number of people who tested negative = $7m = (7 * 12) = 84$

So, the total number of people who tested = $(132+84) = 216$

The number of people who are vaccinated = $6m = (6 * 12) = 72$

The number of people who are not vaccinated = $5m = (5 * 12) = 60$

Now, the ratio of the number of people who tested positive and negative in the central zone = 2:1. Let it be $2a$ and a , respectively.

And, the ratio of the number of people who tested positive and negative in the west zone = 5:4. Let it be $5b$ and $4b$, respectively.

So,

$$5b + 2a = (160 * 2)$$

$$5b + 2a = 320 \dots \dots \dots (I)$$

Also,

$$4b - a = 100 \dots \dots \dots (II)$$

By solving equations (I) and (II), we get,

$$20b + 8a = 1280$$

$$20b - 5a = 500$$

$$13a = 780$$

$$a = 60$$

Now, put the value of a in equation (I), we get,

$$5b + 2a = 320$$

$$5b + (2 * 60) = 320$$

$$5b = 320 - 120$$

$$5b = 200$$

$$b = 40$$

Central Zone

The number of people who tested positive = $2a = (2 * 60) = 120$

The number of people who tested negative = $a = 60$

So, the total number of people who tested = $(120+60) = 180$

West Zone

The number of people who tested positive = $5b = (5 * 40) = 200$

The number of people who tested negative = $4b = (4 * 40) = 160$

So, the total number of people who tested = $(200+160) = 360$

@QuizByMickey

Now,

The total percentage distribution of the total number of people who tested (positive + negative) from all five zones together = 100%

And, the sum of the percentage distribution of the total number of people who tested (positive + negative) from the north, central, and west zone = $\{100 - (24 + 13)\}\% = (100 - 37)\% = 63\%$

The total number of people who tested (positive + negative) from north, central and west zone = $(216 + 180 + 360) = 756$

Now,

$$63\% = 756$$

$$1\% = 12$$

$$100\% = (12 \times 100) = 1200$$

Therefore, the total number of people who tested (positive + negative) from all five zones together = 1200

Now, the total number of people who tested (positive + negative) from east zone = 24% of 1200 = 288

And, the total number of people who tested (positive + negative) from south zone = 13% of 1200 = 156

Answer: C

North Zone

The total number of people who are not vaccinated = 60

Using I

The number of people who died = 10% of 60 = 6

The number of people who are alive = $(60 - 6) = 54$

The required difference = $(54 - 6) = 48$

So, I does not satisfy.

Using II

The number of people who died = 25% of 60 = 15

The number of people who are alive = $(60 - 15) = 45$

The required difference = $(45 - 15) = 30$

So, II does satisfy.

Using III

The number of people who died = 30% of 60 = 18

The number of people who are alive = $(60 - 18) = 42$

The required difference = $(42 - 18) = 24$

So, III does satisfy.

@QuizByMickey

Therefore, both II and III follow.

Hence, the correct answer is option C.

10. Questions

Let the number of people who tested positive and negative in the north zone be $11m$ and $7m$, respectively.

So, the number of people who are vaccinated in the north zone = $6/11$ of $11m = 6m$

And, the number of people who are not vaccinated in the north zone = $5/11$ of $11m = 5m$

Now,

$$7m - 5m = 24$$

$$2m = 24$$

$$m = 12$$

North Zone

The number of people who tested positive = $11m = (11 * 12) = 132$

The number of people who tested negative = $7m = (7 * 12) = 84$

So, the total number of people who tested = $(132+84) = 216$

The number of people who are vaccinated = $6m = (6 * 12) = 72$

The number of people who are not vaccinated = $5m = (5 * 12) = 60$

Now, the ratio of the number of people who tested positive and negative in the central zone = $2:1$. Let it be $2a$ and a , respectively.

And, the ratio of the number of people who tested positive and negative in the west zone = $5:4$. Let it be $5b$ and $4b$, respectively.

So,

$$5b + 2a = (160 * 2)$$

$$5b + 2a = 320 \dots \dots \dots (I)$$

Also,

$$4b - a = 100 \dots \dots \dots (II)$$

By solving equations (I) and (II), we get,

$$20b + 8a = 1280$$

$$20b - 5a = 500$$

$$13a = 780$$

$$a = 60$$

Now, put the value of a in equation (I), we get,

$$5b + 2a = 320$$

@QuizByMickey

$$5b + (2 * 60) = 320$$

$$5b = 320 - 120$$

$$5b = 200$$

$$b = 40$$

Central Zone

$$\text{The number of people who tested positive} = 2a = (2 * 60) = 120$$

$$\text{The number of people who tested negative} = a = 60$$

$$\text{So, the total number of people who tested} = (120+60) = 180$$

West Zone

$$\text{The number of people who tested positive} = 5b = (5 * 40) = 200$$

$$\text{The number of people who tested negative} = 4b = (4 * 40) = 160$$

$$\text{So, the total number of people who tested} = (200+160) = 360$$

Now,

The total percentage distribution of the total number of people who tested (positive + negative) from all five zones together = 100%

And, the sum of the percentage distribution of the total number of people who tested (positive + negative) from the north, central, and west zone = $\{100 - (24+13)\}\% = (100-37)\% = 63\%$

$$\text{The total number of people who tested (positive + negative) from north, central and west zone} = (216+180+360) = 756$$

Now,

$$63\% = 756$$

$$1\% = 12$$

$$100\% = (12 \times 100) = 1200$$

Therefore, the total number of people who tested (positive + negative) from all five zones together = 1200

Now, the total number of people who tested (positive + negative) from east zone = 24% of 1200 = 288

And, the total number of people who tested (positive + negative) from south zone = 13% of 1200 = 156

Answer: A

I: The number of people who tested negative in the north zone = 84

The number of people who tested negative in the central zone = 60

$$\text{The required percentage} = \left[\frac{(84-60)}{60} * 100 \right]\% = \left\{ \frac{(24)}{60} * 100 \right\}\% = 40\%.$$

So, I is definitely true.

II: The total number of people who tested in the central zone = 180

The total number of people who tested in the west zone = 360

The required ratio = 1:2

So, II is definitely not true.

III: The number of people who tested positive in the west zone = 200

So, III is definitely not true.

Therefore, only I is definitely true.

Hence, the correct answer is option A.

11. Questions

Let the number of fours and sixes hit by P be $7a$ and $2a$ respectively.

So, the total number of runs scored by P in fours = $(7a * 4) = 28a$

And, the total number of runs scored by P in sixes = $(2a * 6) = 12a$

Also, let the number of fours and sixes hit by Q be b and b respectively.

So, the total number of runs scored by Q in fours = $(b * 4) = 4b$

And, the total number of runs scored by Q in sixes = $(b * 6) = 6b$

Now,

$$28a - 6b = 36$$

$$14a - 3b = 18 \dots\dots\dots(I)$$

Also,

$$12a - 4b = 4$$

$$3a - b = 1 \dots\dots\dots(II)$$

By solving equation (I) and (II), we get,

$$42a - 9b = 54$$

$$42a - 14b = 14$$

$$5b = 40$$

$$b = 8$$

Now, put the value of b in equation (I), we get,

$$14a - 3b = 18$$

$$14a - (3 * 8) = 18$$

$$14a = 18 + 24$$

$$14a = 42$$

$$a = 3$$

@QuizByMickey

Now,

The total number of runs scored by P in fours = $28a = (28 * 3) = 84$

The total number of runs scored by P in sixes = $12a = (12 * 3) = 36$

So, the total number of runs scored by P in boundaries = $(84+36) = 120$

The total number of runs scored by Q in fours = $4b = (4 * 8) = 32$

The total number of runs scored by Q in sixes = $6b = (6 * 8) = 48$

So, the total number of runs scored by Q in boundaries = $(32+48) = 80$

From the pie chart given above, we have,

The total number of runs scored in boundaries by P = $(Z+20)\%$ of 400 = $(4Z+80)$

So,

$$4Z + 80 = 120$$

$$4Z = 40$$

$$Z = 10$$

The total number of runs scored in boundaries by Q = $Y\%$ of 400 = $4Y$

So,

$$4Y = 80$$

$$Y = 20$$

Now, the sum of the percentage of the number of runs scored in boundaries by P, Q and S together = $\{(Z+20)+Y+Z\}\% = \{(10+20)+20+10\}\% = (30+20+10)\% = 60\%$

The total value of the pie chart = 100%

So, the sum of the percentage of the number of runs scored in boundaries by R and T together = $(100-60)\% = 40\%$

$$\{X+(X+10)\} = 40$$

$$2X = 30$$

$$X = 15$$

@QuizByMickey

Therefore, the percentage of the number of runs scored in boundaries by R = $X\% = 15\%$

And, the percentage of the number of runs scored in boundaries by T = $(X+10)\% = (15+10)\% = 25\%$

For P

The number of runs scored in boundaries = 120

The number of fours hit = $7a = (7 * 3) = 21$

The number of sixes hit = $2a = (2 * 3) = 6$

The total number of runs scored in fours = $28a = (28 * 3) = 84$

The total number of runs scored in sixes = $12a = (12 * 3) = 36$

For Q

The number of runs scored in boundaries = 80

The number of fours hit = $b = 8$

The number of sixes hit = $b = 8$

The total number of runs scored in fours = $4b = (4 * 8) = 32$

The total number of runs scored in sixes = $6b = (6 * 8) = 48$

For R

The number of runs scored in boundaries = $X\%$ of 400 = 15% of 400 = 60

Let the number of fours and sixes hit be $6c$ and c , respectively.

So, the total number of runs scored in fours = $(6c * 4) = 24c$

And, the total number of runs scored in sixes = $(c * 6) = 6c$

Now,

$$24c + 6c = 60$$

$$30c = 60$$

$$c = 2$$

Therefore,

The number of fours hit = $6c = (6 * 2) = 12$

The number of sixes hit = $c = 2$

The total number of runs scored in fours = $24c = (24 * 2) = 48$

The total number of runs scored in sixes = $6c = (6 * 2) = 12$

For S

The number of runs scored in boundaries = $Z\%$ of 400 = 10% of 400 = 40

Let the number of fours and sixes hit be $7d$ and $2d$, respectively.

So, the total number of runs scored in fours = $(7d * 4) = 28d$

And, the total number of runs scored in sixes = $(2d * 6) = 12d$

Now,

$$28d + 12d = 40$$

$$40d = 40$$

$$d = 1$$

Therefore,

@QuizByMickey

The number of fours hit = $7d = (7 * 1) = 7$

The number of sixes hit = $2d = (2 * 1) = 2$

The total number of runs scored in fours = $28d = (28 * 1) = 28$

The total number of runs scored in sixes = $12d = (12 * 1) = 12$

For T

The number of runs scored in boundaries = $(X+10)\%$ of 400 = $(15+10)\%$ of 400 = 25% of 400 = 100

Let the number of fours and sixes hit be $19e$ and $4e$, respectively.

So, the total number of runs scored in fours = $(19e * 4) = 76e$

And, the total number of runs scored in sixes = $(4e * 6) = 24e$

Now,

$$76e + 24e = 100$$

$$100e = 100$$

$$e = 1$$

Therefore,

The number of fours hit = $19e$

The number of sixes hit = $4e$

The total number of runs scored in fours = $76e = (76 * 1) = 76$

The total number of runs scored in sixes = $24e = (24 * 1) = 24$

Name of the players	The number of fours hit	The number of sixes hit	The total number of runs scored in fours	The total number of runs scored in sixes	The total number of runs scored in boundaries (fours + sixes)
P	21	6	84	36	120
Q	8	8	32	48	80
R	12	2	48	12	60
S	7	2	28	12	40
T	19	4	76	24	100
Total	67	22	268	132	400

Answer: A

According to the question,

For P

The number of runs scored in boundaries = 120

The number of runs scored other than boundaries = K

So, the total number of runs scored = $(120+K)$

The total number of balls faced = 240

Now,

$$\{(120+K)/240\} * 100 = (K-160)$$

$$(120+K)/240 = (K-160)/100$$

$$12000 + 100K = 240K - 38400$$

$$140K = 50400$$

$$K = 360$$

Therefore, the total number of runs scored by P = $(120+K) = (120+360) = 480$

So, the required percentage = $(120/480) * 100 = 25\%$.

Hence, the correct answer is option A.

12. Questions

Let the number of fours and sixes hit by P be 7a and 2a respectively.

So, the total number of runs scored by P in fours = $(7a * 4) = 28a$

And, the total number of runs scored by P in sixes = $(2a * 6) = 12a$

Also, let the number of fours and sixes hit by Q be b and b respectively.

So, the total number of runs scored by Q in fours = $(b * 4) = 4b$

And, the total number of runs scored by Q in sixes = $(b * 6) = 6b$

Now,

$$28a - 6b = 36$$

$$14a - 3b = 18 \dots\dots\dots(I)$$

Also,

$$12a - 4b = 4$$

$$3a - b = 1 \dots\dots\dots(II)$$

By solving equation (I) and (II), we get,

$$42a - 9b = 54$$

$$42a - 14b = 14$$

$$5b = 40$$

$$b = 8$$

Now, put the value of b in equation (I), we get,

@QuizByMickey

$$14a - 3b = 18$$

$$14a - (3 * 8) = 18$$

$$14a = 18 + 24$$

$$14a = 42$$

$$a = 3$$

Now,

$$\text{The total number of runs scored by P in fours} = 28a = (28 * 3) = 84$$

$$\text{The total number of runs scored by P in sixes} = 12a = (12 * 3) = 36$$

$$\text{So, the total number of runs scored by P in boundaries} = (84+36) = 120$$

$$\text{The total number of runs scored by Q in fours} = 4b = (4 * 8) = 32$$

$$\text{The total number of runs scored by Q in sixes} = 6b = (6 * 8) = 48$$

$$\text{So, the total number of runs scored by Q in boundaries} = (32+48) = 80$$

From the pie chart given above, we have,

$$\text{The total number of runs scored in boundaries by P} = (Z+20)\% \text{ of } 400 = (4Z+80)$$

So,

$$4Z + 80 = 120$$

$$4Z = 40$$

$$Z = 10$$

$$\text{The total number of runs scored in boundaries by Q} = Y\% \text{ of } 400 = 4Y$$

So,

$$4Y = 80$$

$$Y = 20$$

Now, the sum of the percentage of the number of runs scored in boundaries by P, Q and S together =
 $\{(Z+20)+Y+Z\}\% = \{(10+20)+20+10\}\% = (30+20+10)\% = 60\%$

The total value of the pie chart = 100%

@QuizByMickey

So, the sum of the percentage of the number of runs scored in boundaries by R and T together =
 $(100-60)\% = 40\%$

$$\{X+(X+10)\} = 40$$

$$2X = 30$$

$$X = 15$$

Therefore, the percentage of the number of runs scored in boundaries by R = $X\% = 15\%$

And, the percentage of the number of runs scored in boundaries by T = $(X+10)\% = (15+10)\% = 25\%$

For P

The number of runs scored in boundaries = 120

The number of fours hit = $7a = (7 * 3) = 21$

The number of sixes hit = $2a = (2 * 3) = 6$

The total number of runs scored in fours = $28a = (28 * 3) = 84$

The total number of runs scored in sixes = $12a = (12 * 3) = 36$

For Q

The number of runs scored in boundaries = 80

The number of fours hit = $b = 8$

The number of sixes hit = $b = 8$

The total number of runs scored in fours = $4b = (4 * 8) = 32$

The total number of runs scored in sixes = $6b = (6 * 8) = 48$

For R

The number of runs scored in boundaries = $X\%$ of 400 = 15% of 400 = 60

Let the number of fours and sixes hit be $6c$ and c , respectively.

So, the total number of runs scored in fours = $(6c * 4) = 24c$

And, the total number of runs scored in sixes = $(c * 6) = 6c$

Now,

$$24c + 6c = 60$$

$$30c = 60$$

$$c = 2$$

Therefore,

The number of fours hit = $6c = (6 * 2) = 12$

The number of sixes hit = $c = 2$

The total number of runs scored in fours = $24c = (24 * 2) = 48$

The total number of runs scored in sixes = $6c = (6 * 2) = 12$

For S

The number of runs scored in boundaries = $Z\%$ of 400 = 10% of 400 = 40

Let the number of fours and sixes hit be $7d$ and $2d$, respectively.

So, the total number of runs scored in fours = $(7d * 4) = 28d$

And, the total number of runs scored in sixes = $(2d * 6) = 12d$

@QuizByMickey

Now,

$$28d + 12d = 40$$

$$40d = 40$$

$$d = 1$$

Therefore,

$$\text{The number of fours hit} = 7d = (7 * 1) = 7$$

$$\text{The number of sixes hit} = 2d = (2 * 1) = 2$$

$$\text{The total number of runs scored in fours} = 28d = (28 * 1) = 28$$

$$\text{The total number of runs scored in sixes} = 12d = (12 * 1) = 12$$

For T

$$\text{The number of runs scored in boundaries} = (X+10)\% \text{ of } 400 = (15+10)\% \text{ of } 400 = 25\% \text{ of } 400 = 100$$

Let the number of fours and sixes hit be $19e$ and $4e$, respectively.

$$\text{So, the total number of runs scored in fours} = (19e * 4) = 76e$$

$$\text{And, the total number of runs scored in sixes} = (4e * 6) = 24e$$

Now,

$$76e + 24e = 100$$

$$100e = 100$$

$$e = 1$$

Therefore,

$$\text{The number of fours hit} = 19e$$

$$\text{The number of sixes hit} = 4e$$

$$\text{The total number of runs scored in fours} = 76e = (76 * 1) = 76$$

$$\text{The total number of runs scored in sixes} = 24e = (24 * 1) = 24$$

@QuizByMickey

Name of the players	The number of fours hit	The number of sixes hit	The total number of runs scored in fours	The total number of runs scored in sixes	The total number of runs scored in boundaries (fours + sixes)
P	21	6	84	36	120
Q	8	8	32	48	80
R	12	2	48	12	60
S	7	2	28	12	40
T	19	4	76	24	100
Total	67	22	268	132	400

Answer: D

For Q

The number of runs scored in boundaries = 80

The strike-rate = 80

Let the total number of balls faced be N.

Using I

The number of runs scored other than boundaries = $\{(80/20) * 80\} = 320$

Now, the total number of runs scored = $(80+320) = 400$

So,

$$\{(400/N) * 100\} = 80$$

$$400/N = 80/100$$

$$400/N = 4/5$$

$$4N = 2000$$

$$N = 500$$

So, I does satisfy.

@QuizByMickey

Using II

The number of runs scored other than boundaries = $\{(80/25) * 75\} = 240$

Now, the total number of runs scored = $(80+240) = 320$

So,

$$\{(320/N) * 100\} = 80$$

$$320/N = 80/100$$

$$320/N = 4/5$$

$$4N = 1600$$

$$N = 400$$

So, II does satisfy.

Using III

$$\text{The number of runs scored other than boundaries} = \{(80/40) * 60\} = 120$$

$$\text{Now, the total number of runs scored} = (80+120) = 200$$

So,

$$\{(200/N) * 100\} = 80$$

$$200/N = 80/100$$

$$200/N = 4/5$$

$$4N = 1000$$

$$N = 250$$

So, III does satisfy.

Therefore, all I, II, and III follow.

Hence, the correct answer is option D.

13. Questions

Let the number of fours and sixes hit by P be $7a$ and $2a$ respectively.

$$\text{So, the total number of runs scored by P in fours} = (7a * 4) = 28a$$

$$\text{And, the total number of runs scored by P in sixes} = (2a * 6) = 12a$$

Also, let the number of fours and sixes hit by Q be b and b respectively.

$$\text{So, the total number of runs scored by Q in fours} = (b * 4) = 4b$$

$$\text{And, the total number of runs scored by Q in sixes} = (b * 6) = 6b$$

Now,

$$28a - 6b = 36$$

$$14a - 3b = 18 \dots\dots\dots(I)$$

Also,

$$12a - 4b = 4$$

$$3a - b = 1 \dots\dots\dots(II)$$

By solving equation (I) and (II), we get,

$$42a - 9b = 54$$

$$42a - 14b = 14$$

@QuizByMickey

$$5b = 40$$

$$b = 8$$

Now, put the value of b in equation (I), we get,

$$14a - 3b = 18$$

$$14a - (3 * 8) = 18$$

$$14a = 18 + 24$$

$$14a = 42$$

$$a = 3$$

Now,

$$\text{The total number of runs scored by P in fours} = 28a = (28 * 3) = 84$$

$$\text{The total number of runs scored by P in sixes} = 12a = (12 * 3) = 36$$

$$\text{So, the total number of runs scored by P in boundaries} = (84+36) = 120$$

$$\text{The total number of runs scored by Q in fours} = 4b = (4 * 8) = 32$$

$$\text{The total number of runs scored by Q in sixes} = 6b = (6 * 8) = 48$$

$$\text{So, the total number of runs scored by Q in boundaries} = (32+48) = 80$$

From the pie chart given above, we have,

$$\text{The total number of runs scored in boundaries by P} = (Z+20)\% \text{ of } 400 = (4Z+80)$$

So,

$$4Z + 80 = 120$$

$$4Z = 40$$

$$Z = 10$$

$$\text{The total number of runs scored in boundaries by Q} = Y\% \text{ of } 400 = 4Y$$

So,

$$4Y = 80$$

$$Y = 20$$

@QuizByMickey

$$\text{Now, the sum of the percentage of the number of runs scored in boundaries by P, Q and S together} = \{(Z+20)+Y+Z\}\% = \{(10+20)+20+10\}\% = (30+20+10)\% = 60\%$$

$$\text{The total value of the pie chart} = 100\%$$

$$\text{So, the sum of the percentage of the number of runs scored in boundaries by R and T together} = (100-60)\% = 40\%$$

$$\{X+(X+10)\} = 40$$

$$2X = 30$$

$$X = 15$$

Therefore, the percentage of the number of runs scored in boundaries by R = $X\% = 15\%$

And, the percentage of the number of runs scored in boundaries by T = $(X+10)\% = (15+10)\% = 25\%$

For P

The number of runs scored in boundaries = 120

The number of fours hit = $7a = (7 * 3) = 21$

The number of sixes hit = $2a = (2 * 3) = 6$

The total number of runs scored in fours = $28a = (28 * 3) = 84$

The total number of runs scored in sixes = $12a = (12 * 3) = 36$

For Q

The number of runs scored in boundaries = 80

The number of fours hit = $b = 8$

The number of sixes hit = $b = 8$

The total number of runs scored in fours = $4b = (4 * 8) = 32$

The total number of runs scored in sixes = $6b = (6 * 8) = 48$

For R

The number of runs scored in boundaries = $X\%$ of 400 = 15% of 400 = 60

Let the number of fours and sixes hit be $6c$ and c , respectively.

So, the total number of runs scored in fours = $(6c * 4) = 24c$

And, the total number of runs scored in sixes = $(c * 6) = 6c$

Now,

$$24c + 6c = 60$$

$$30c = 60$$

$$c = 2$$

Therefore,

The number of fours hit = $6c = (6 * 2) = 12$

The number of sixes hit = $c = 2$

The total number of runs scored in fours = $24c = (24 * 2) = 48$

The total number of runs scored in sixes = $6c = (6 * 2) = 12$

For S

The number of runs scored in boundaries = $Z\%$ of 400 = 10% of 400 = 40

@QuizByMickey

Let the number of fours and sixes hit be $7d$ and $2d$, respectively.

So, the total number of runs scored in fours = $(7d * 4) = 28d$

And, the total number of runs scored in sixes = $(2d * 6) = 12d$

Now,

$$28d + 12d = 40$$

$$40d = 40$$

$$d = 1$$

Therefore,

The number of fours hit = $7d = (7 * 1) = 7$

The number of sixes hit = $2d = (2 * 1) = 2$

The total number of runs scored in fours = $28d = (28 * 1) = 28$

The total number of runs scored in sixes = $12d = (12 * 1) = 12$

For T

The number of runs scored in boundaries = $(X+10)\%$ of 400 = $(15+10)\%$ of 400 = 25% of 400 = 100

Let the number of fours and sixes hit be $19e$ and $4e$, respectively.

So, the total number of runs scored in fours = $(19e * 4) = 76e$

And, the total number of runs scored in sixes = $(4e * 6) = 24e$

Now,

$$76e + 24e = 100$$

$$100e = 100$$

$$e = 1$$

Therefore,

The number of fours hit = $19e$

The number of sixes hit = $4e$

The total number of runs scored in fours = $76e = (76 * 1) = 76$

The total number of runs scored in sixes = $24e = (24 * 1) = 24$

@QuizByMickey

Name of the players	The number of fours hit	The number of sixes hit	The total number of runs scored in fours	The total number of runs scored in sixes	The total number of runs scored in boundaries (fours + sixes)
P	21	6	84	36	120
Q	8	8	32	48	80
R	12	2	48	12	60
S	7	2	28	12	40
T	19	4	76	24	100
Total	67	22	268	132	400

Answer: A

For R

Let the number of singles, doubles, and triples run by R be $5n$, $10n$, and n , respectively.

So, the total number of runs scored in singles = $(5n * 1) = 5n$

The total number of runs scored in doubles = $(10n * 2) = 20n$

The total number of runs scored in triple = $(n * 3) = 3n$

So, the total number of runs scored other than boundaries = $(5n+20n+3n) = 28n$

Also, the total number of runs scored in boundaries = 60

Now, the total number of runs scored = $(28n+60)$

And, the total number of balls faced = 160

So,

$$\{(28n+60)/160\} * 100 = 125$$

$$(28n+60)/160 = 125/100$$

$$(28n+60)/160 = 5/4$$

$$112n + 240 = 800$$

$$112n = 560$$

$$n = 5$$

Therefore, the number of singles scored = $5n = (5 * 5) = 25$

The number of doubles scored = $10n = (10 * 5) = 50$

The number of triples scored = $n = 5$

So, the total number of balls faced to score the number of runs other than boundaries = $(25+50+5) = 80$

Also, the number of fours scored = 12

@QuizByMickey

And, the number of sixes scored = 2

So, the total number of balls faced to hit boundaries = $(12 + 2) = 14$

Now, the total number of balls faced to score the total number of runs = $(80 + 14) = 94$

Also, the total number of balls faced = 160

Therefore, the number of dot balls faced = $(160 - 94) = 66$.

Hence, the correct answer is option A.

14. Questions

Let the number of fours and sixes hit by P be $7a$ and $2a$ respectively.

So, the total number of runs scored by P in fours = $(7a * 4) = 28a$

And, the total number of runs scored by P in sixes = $(2a * 6) = 12a$

Also, let the number of fours and sixes hit by Q be b and b respectively.

So, the total number of runs scored by Q in fours = $(b * 4) = 4b$

And, the total number of runs scored by Q in sixes = $(b * 6) = 6b$

Now,

$$28a - 6b = 36$$

$$14a - 3b = 18 \dots\dots\dots(I)$$

Also,

$$12a - 4b = 4$$

$$3a - b = 1 \dots\dots\dots(II)$$

By solving equation (I) and (II), we get,

$$42a - 9b = 54$$

$$42a - 14b = 14$$

$$5b = 40$$

$$b = 8$$

Now, put the value of b in equation (I), we get,

$$14a - 3b = 18$$

$$14a - (3 * 8) = 18$$

$$14a = 18 + 24$$

$$14a = 42$$

$$a = 3$$

Now,

@QuizByMickey

The total number of runs scored by P in fours = $28a = (28 * 3) = 84$

The total number of runs scored by P in sixes = $12a = (12 * 3) = 36$

So, the total number of runs scored by P in boundaries = $(84+36) = 120$

The total number of runs scored by Q in fours = $4b = (4 * 8) = 32$

The total number of runs scored by Q in sixes = $6b = (6 * 8) = 48$

So, the total number of runs scored by Q in boundaries = $(32+48) = 80$

From the pie chart given above, we have,

The total number of runs scored in boundaries by P = $(Z+20)\%$ of 400 = $(4Z+80)$

So,

$$4Z + 80 = 120$$

$$4Z = 40$$

$$Z = 10$$

The total number of runs scored in boundaries by Q = $Y\%$ of 400 = $4Y$

So,

$$4Y = 80$$

$$Y = 20$$

Now, the sum of the percentage of the number of runs scored in boundaries by P, Q and S together = $\{(Z+20)+Y+Z\}\% = \{(10+20)+20+10\}\% = (30+20+10)\% = 60\%$

The total value of the pie chart = 100%

So, the sum of the percentage of the number of runs scored in boundaries by R and T together = $(100-60)\% = 40\%$

$$\{X+(X+10)\} = 40$$

$$2X = 30$$

$$X = 15$$

Therefore, the percentage of the number of runs scored in boundaries by R = $X\% = 15\%$

And, the percentage of the number of runs scored in boundaries by T = $(X+10)\% = (15+10)\% = 25\%$

For P

The number of runs scored in boundaries = 120

The number of fours hit = $7a = (7 * 3) = 21$

The number of sixes hit = $2a = (2 * 3) = 6$

The total number of runs scored in fours = $28a = (28 * 3) = 84$

The total number of runs scored in sixes = $12a = (12 * 3) = 36$

For Q

The number of runs scored in boundaries = 80

The number of fours hit = $b = 8$

The number of sixes hit = $b = 8$

The total number of runs scored in fours = $4b = (4 * 8) = 32$

The total number of runs scored in sixes = $6b = (6 * 8) = 48$

For R

The number of runs scored in boundaries = $X\%$ of 400 = 15% of 400 = 60

Let the number of fours and sixes hit be $6c$ and c , respectively.

So, the total number of runs scored in fours = $(6c * 4) = 24c$

And, the total number of runs scored in sixes = $(c * 6) = 6c$

Now,

$$24c + 6c = 60$$

$$30c = 60$$

$$c = 2$$

Therefore,

The number of fours hit = $6c = (6 * 2) = 12$

The number of sixes hit = $c = 2$

The total number of runs scored in fours = $24c = (24 * 2) = 48$

The total number of runs scored in sixes = $6c = (6 * 2) = 12$

For S

The number of runs scored in boundaries = $Z\%$ of 400 = 10% of 400 = 40

Let the number of fours and sixes hit be $7d$ and $2d$, respectively.

So, the total number of runs scored in fours = $(7d * 4) = 28d$

And, the total number of runs scored in sixes = $(2d * 6) = 12d$

Now,

$$28d + 12d = 40$$

$$40d = 40$$

$$d = 1$$

Therefore,

The number of fours hit = $7d = (7 * 1) = 7$

@QuizByMickey

The number of sixes hit = $2d = (2 * 1) = 2$

The total number of runs scored in fours = $28d = (28 * 1) = 28$

The total number of runs scored in sixes = $12d = (12 * 1) = 12$

For T

The number of runs scored in boundaries = $(X+10)\%$ of 400 = $(15+10)\%$ of 400 = 25% of 400 = 100

Let the number of fours and sixes hit be $19e$ and $4e$, respectively.

So, the total number of runs scored in fours = $(19e * 4) = 76e$

And, the total number of runs scored in sixes = $(4e * 6) = 24e$

Now,

$$76e + 24e = 100$$

$$100e = 100$$

$$e = 1$$

Therefore,

The number of fours hit = $19e$

The number of sixes hit = $4e$

The total number of runs scored in fours = $76e = (76 * 1) = 76$

The total number of runs scored in sixes = $24e = (24 * 1) = 24$

Name of the players	The number of fours hit	The number of sixes hit	The total number of runs scored in fours	The total number of runs scored in sixes	The total number of runs scored in boundaries (fours + sixes)
P	21	6	84	36	120
Q	8	8	32	48	80
R	12	2	48	12	60
S	7	2	28	12	40
T	19	4	76	24	100
Total	67	22	268	132	400

Answer: B

The number of runs scored in boundaries by,

$$S = 40$$

$$T = 100$$

So, the number of runs scored other than boundaries by,

$$S = \{(40/2) * 3\} = 60$$

$$T = \{(100/20) * 80\} = 400$$

Now, the total number of runs scored by,

$$S = (40 + 60) = 100$$

$$T = (100 + 400) = 500$$

Therefore, the total number of runs scored by S and T together = $(100 + 500) = 600$

$$\text{So, } L = 600$$

600 is exactly divisible by only 12 among the given options.

So, the question mark = 12.

Hence, the correct answer is option B.

15. Questions

Let the number of fours and sixes hit by P be $7a$ and $2a$ respectively.

So, the total number of runs scored by P in fours = $(7a * 4) = 28a$

And, the total number of runs scored by P in sixes = $(2a * 6) = 12a$

Also, let the number of fours and sixes hit by Q be b and b respectively.

So, the total number of runs scored by Q in fours = $(b * 4) = 4b$

And, the total number of runs scored by Q in sixes = $(b * 6) = 6b$

Now,

$$28a - 6b = 36$$

$$14a - 3b = 18 \dots\dots\dots(I)$$

Also,

$$12a - 4b = 4$$

$$3a - b = 1 \dots\dots\dots(II)$$

By solving equation (I) and (II), we get,

$$42a - 9b = 54$$

$$42a - 14b = 14$$

$$5b = 40$$

$$b = 8$$

Now, put the value of b in equation (I), we get,

$$14a - 3b = 18$$

$$14a - (3 * 8) = 18$$

@QuizByMickey

$$14a = 18 + 24$$

$$14a = 42$$

$$a = 3$$

Now,

$$\text{The total number of runs scored by P in fours} = 28a = (28 * 3) = 84$$

$$\text{The total number of runs scored by P in sixes} = 12a = (12 * 3) = 36$$

$$\text{So, the total number of runs scored by P in boundaries} = (84+36) = 120$$

$$\text{The total number of runs scored by Q in fours} = 4b = (4 * 8) = 32$$

$$\text{The total number of runs scored by Q in sixes} = 6b = (6 * 8) = 48$$

$$\text{So, the total number of runs scored by Q in boundaries} = (32+48) = 80$$

From the pie chart given above, we have,

$$\text{The total number of runs scored in boundaries by P} = (Z+20)\% \text{ of } 400 = (4Z+80)$$

So,

$$4Z + 80 = 120$$

$$4Z = 40$$

$$Z = 10$$

$$\text{The total number of runs scored in boundaries by Q} = Y\% \text{ of } 400 = 4Y$$

So,

$$4Y = 80$$

$$Y = 20$$

$$\text{Now, the sum of the percentage of the number of runs scored in boundaries by P, Q and S together} = \{(Z+20)+Y+Z\}\% = \{(10+20)+20+10\}\% = (30+20+10)\% = 60\%$$

$$\text{The total value of the pie chart} = 100\%$$

$$\text{So, the sum of the percentage of the number of runs scored in boundaries by R and T together} = (100-60)\% = 40\%$$

$$\{X+(X+10)\} = 40$$

$$2X = 30$$

$$X = 15$$

$$\text{Therefore, the percentage of the number of runs scored in boundaries by R} = X\% = 15\%$$

$$\text{And, the percentage of the number of runs scored in boundaries by T} = (X+10)\% = (15+10)\% = 25\%$$

For P

$$\text{The number of runs scored in boundaries} = 120$$

@QuizByMickey

The number of fours hit = $7a = (7 * 3) = 21$

The number of sixes hit = $2a = (2 * 3) = 6$

The total number of runs scored in fours = $28a = (28 * 3) = 84$

The total number of runs scored in sixes = $12a = (12 * 3) = 36$

For Q

The number of runs scored in boundaries = 80

The number of fours hit = $b = 8$

The number of sixes hit = $b = 8$

The total number of runs scored in fours = $4b = (4 * 8) = 32$

The total number of runs scored in sixes = $6b = (6 * 8) = 48$

For R

The number of runs scored in boundaries = $X\%$ of 400 = 15% of 400 = 60

Let the number of fours and sixes hit be $6c$ and c , respectively.

So, the total number of runs scored in fours = $(6c * 4) = 24c$

And, the total number of runs scored in sixes = $(c * 6) = 6c$

Now,

$$24c + 6c = 60$$

$$30c = 60$$

$$c = 2$$

Therefore,

The number of fours hit = $6c = (6 * 2) = 12$

The number of sixes hit = $c = 2$

The total number of runs scored in fours = $24c = (24 * 2) = 48$

The total number of runs scored in sixes = $6c = (6 * 2) = 12$

For S

The number of runs scored in boundaries = $Z\%$ of 400 = 10% of 400 = 40

Let the number of fours and sixes hit be $7d$ and $2d$, respectively.

So, the total number of runs scored in fours = $(7d * 4) = 28d$

And, the total number of runs scored in sixes = $(2d * 6) = 12d$

Now,

$$28d + 12d = 40$$

@QuizByMickey

$$40d = 40$$

$$d = 1$$

Therefore,

$$\text{The number of fours hit} = 7d = (7 * 1) = 7$$

$$\text{The number of sixes hit} = 2d = (2 * 1) = 2$$

$$\text{The total number of runs scored in fours} = 28d = (28 * 1) = 28$$

$$\text{The total number of runs scored in sixes} = 12d = (12 * 1) = 12$$

For T

$$\text{The number of runs scored in boundaries} = (X+10)\% \text{ of } 400 = (15+10)\% \text{ of } 400 = 25\% \text{ of } 400 = 100$$

Let the number of fours and sixes hit be $19e$ and $4e$, respectively.

$$\text{So, the total number of runs scored in fours} = (19e * 4) = 76e$$

$$\text{And, the total number of runs scored in sixes} = (4e * 6) = 24e$$

Now,

$$76e + 24e = 100$$

$$100e = 100$$

$$e = 1$$

Therefore,

$$\text{The number of fours hit} = 19e$$

$$\text{The number of sixes hit} = 4e$$

$$\text{The total number of runs scored in fours} = 76e = (76 * 1) = 76$$

$$\text{The total number of runs scored in sixes} = 24e = (24 * 1) = 24$$

Name of the players	The number of fours hit	The number of sixes hit	The total number of runs scored in fours	The total number of runs scored in sixes	The total number of runs scored in boundaries (fours + sixes)
P	21	6	84	36	120
Q	8	8	32	48	80
R	12	2	48	12	60
S	7	2	28	12	40
T	19	4	76	24	100
Total	67	22	268	132	400

Answer: B

For U

The total number of runs scored in fours = $(23 * 4) = 92$

And, the total number of runs scored in sixes = $(8 * 6) = 48$

So, the total number of runs scored in boundaries = $(92 + 48) = 140$

Also, the total number of runs scored (boundaries + other than boundaries) = $28X = (28 * 15) = 420$ (**X = 15, from common explanation**)

Therefore, the number of runs scored other than boundaries = $(420 - 140) = 280$

So, the ratio of the number of runs scored in boundaries and other than boundaries = $140:280 = 1:2$

Therefore, $F = 1$

And, $G = 2$

So, $F < G$.

Hence, the correct answer is option B.

16. Questions

Let the total number of books published in January and February be $4m$ and $5m$, respectively.

January

The total number of books published = $4m$

The number of available books = 75% of $4m = 3m$

The number of unavailable books = $(4m - 3m) = m$

The number of available books unsold = 120

The number of available books sold = $(3m - 120)$

February

The total number of books published = $5m$

The number of available books = 80% of $5m = 4m$

The number of unavailable books = $(5m - 4m) = m$

The number of available books unsold = 80

The number of available books sold = $(4m - 80)$

Now,

$$(4m - 80) - (3m - 120) = 140$$

$$4m - 220 = 3m - 120$$

$$m = 100$$

Now,

@QuizByMickey

The ratio of the total number of books published in March to April = $100:150 = 2:3$. Let it be $2n$ and $3n$ respectively.

March

The total number of books published = $2n$

The number of available books = 50% of $2n = n$

The number of unavailable books = $(2n - n) = n$

The number of available books unsold = 50

The number of available books sold = $(n - 50)$

April

The total number of books published = $3n$

The number of available books = 60% of $3n = 1.80n$

The number of unavailable books = $(3n - 1.80n) = 1.20n$

The number of available books unsold = 72

The number of available books sold = $(1.80n - 72)$

Now,

$$(1.80n - 72) - (n - 50) = 74$$

$$1.80n - 146 = n - 50$$

$$0.80n = 96$$

$$n = 120$$

January

The total number of books published = $4m = (4 * 100) = 400$

The number of available books = $3m = (3 * 100) = 300$

The number of unavailable books = $m = 100$

The number of available books sold = $(3m - 120) = (300 - 120) = 180$

February

The total number of books published = $5m = (5 * 100) = 500$

The number of available books = $4m = (4 * 100) = 400$

The number of unavailable books = $m = 100$

The number of available books sold = $(4m - 80) = (400 - 80) = 320$

March

The total number of books published = $2n = (2 * 120) = 240$

@QuizByMickey

The number of available books = $n = 120$

The number of unavailable books = $n = 120$

The number of available books sold = $(n-50) = (120-50) = 70$

April

The total number of books published = $3n = (3 * 120) = 360$

The number of available books = $1.80n = (1.80 * 120) = 216$

The number of unavailable books = $1.20n = (1.20 * 120) = 144$

The number of available books sold = $(1.80n-72) = (216-72) = 144$

May

Let the total number of books published be k .

So, the number of available books = 45% of $k = 0.45k$

And the number of unavailable books = $(k-0.45k) = 0.55k$

The number of available books unsold = 36

So, the number of available books sold = $(0.45k-36)$

Now,

$$(0.45k-36)/0.55k = 5/11$$

$$4.95k - 396 = 2.75k$$

$$2.20k = 396$$

$$k = 180$$

Therefore,

The total number of books published = $k = 180$

The number of available books = $0.45k = (0.45 * 180) = 81$

The number of unavailable books = $0.55k = (0.55 * 180) = 99$

The number of available books sold = $(0.45k-36) = (81-36) = 45$

@QuizByMickey

Month	The total number of books published	The number of available books	The number of unavailable books	The number of available books sold	The number of available books unsold
January	400	300	100	180	120
February	500	400	100	320	80
March	240	120	120	70	50
April	360	216	144	144	72
May	180	81	99	45	36
Total	1680	1117	563	759	358

Answer: B

According to the question,

February

The total number of books published = 500

The number of non-fictional books published = 200

So, the number of fictional books published = $(500 - 200) = 300$

January

The total number of books published = 400

The number of fictional books published = $\{(300/15) * 14\} = 280$

So, the number of non-fictional books published = $(400 - 280) = 120$

Therefore, A = $\{(120/400) * 100\} = 30$

And, B = $(200 - 120) = 80$

I: $(A + 50) = B$

$(30 + 50) = 80$

$80 = 80$

So, I is correct.

II: $(B/2) = (A + 10)$

$(80/2) = (30 + 10)$

$40 = 40$

So, II is correct.

III: $(A + B) = 120$

@QuizByMickey

$$(30+80) = 110$$

So, III is not correct.

Therefore, both equations I and II follow.

Hence, the correct answer is option B

17. Questions

Let the total number of books published in January and February be $4m$ and $5m$, respectively.

January

The total number of books published = $4m$

The number of available books = 75% of $4m = 3m$

The number of unavailable books = $(4m-3m) = m$

The number of available books unsold = 120

The number of available books sold = $(3m-120)$

February

The total number of books published = $5m$

The number of available books = 80% of $5m = 4m$

The number of unavailable books = $(5m-4m) = m$

The number of available books unsold = 80

The number of available books sold = $(4m-80)$

Now,

$$(4m-80) - (3m-120) = 140$$

$$4m - 220 = 3m - 120$$

$$m = 100$$

Now,

The ratio of the total number of books published in March to April = $100:150 = 2:3$. Let it be $2n$ and $3n$ respectively.

March

The total number of books published = $2n$

The number of available books = 50% of $2n = n$

The number of unavailable books = $(2n-n) = n$

@QuizByMickey

The number of available books unsold = 50

The number of available books sold = $(n-50)$

April

The total number of books published = $3n$

The number of available books = 60% of $3n = 1.80n$

The number of unavailable books = $(3n-1.80n) = 1.20n$

The number of available books unsold = 72

The number of available books sold = $(1.80n-72)$

Now,

$$(1.80n-72) - (n-50) = 74$$

$$1.80n - 146 = n - 50$$

$$0.80n = 96$$

$$n = 120$$

January

The total number of books published = $4m = (4 * 100) = 400$

The number of available books = $3m = (3 * 100) = 300$

The number of unavailable books = $m = 100$

The number of available books sold = $(3m-120) = (300-120) = 180$

February

The total number of books published = $5m = (5 * 100) = 500$

The number of available books = $4m = (4 * 100) = 400$

The number of unavailable books = $m = 100$

The number of available books sold = $(4m-80) = (400-80) = 320$

March

The total number of books published = $2n = (2 * 120) = 240$

The number of available books = $n = 120$

The number of unavailable books = $n = 120$

The number of available books sold = $(n-50) = (120-50) = 70$

April

The total number of books published = $3n = (3 * 120) = 360$

The number of available books = $1.80n = (1.80 * 120) = 216$

@QuizByMickey

The number of unavailable books = $1.20n = (1.20 * 120) = 144$

The number of available books sold = $(1.80n-72) = (216-72) = 144$

May

Let the total number of books published be k.

So, the number of available books = 45% of k = $0.45k$

And the number of unavailable books = $(k-0.45k) = 0.55k$

The number of available books unsold = 36

So, the number of available books sold = $(0.45k-36)$

Now,

$$(0.45k-36)/0.55k = 5/11$$

$$4.95k - 396 = 2.75k$$

$$2.20k = 396$$

$$k = 180$$

Therefore,

The total number of books published = k = 180

The number of available books = $0.45k = (0.45 * 180) = 81$

The number of unavailable books = $0.55k = (0.55 * 180) = 99$

The number of available books sold = $(0.45k-36) = (81-36) = 45$

Month	The total number of books published	The number of available books	The number of unavailable books	The number of available books sold	The number of available books unsold
January	400	300	100	180	120
February	500	400	100	320	80
March	240	120	120	70	50
April	360	216	144	144	72
May	180	81	99	45	36
Total	1680	1117	563	759	358

Answer: A

March

The total number of available books sold = 70

And, the S.P. of each book sold = $\{50+(X\% \text{ of } 50)\} = \text{Rs. } (50+0.50X)$

So, the overall S.P. of all books sold = $\{70 * (50 + 0.50X)\} = \text{Rs. } 5600$

$$50 + 0.5X = 80$$

$$0.5X = 30$$

$$X = 60$$

$$\text{Now, } Y = \{(X/3) + 50\} = \{(60/3) + 50\} = (20 + 50) = 70$$

Therefore, the value of $[\sqrt{(5X - 3Y + 31)}] = [\sqrt{(5 * 60 - 3 * 70 + 31)}] = \sqrt{(300 - 210 + 31)} = \sqrt{(300 - 179)} = \sqrt{121} = 11$.

Hence, the correct answer is option A.

18. Questions

Let the total number of books published in January and February be 4m and 5m, respectively.

January

The total number of books published = 4m

The number of available books = 75% of 4m = 3m

The number of unavailable books = (4m - 3m) = m

The number of available books unsold = 120

The number of available books sold = (3m - 120)

February

The total number of books published = 5m

The number of available books = 80% of 5m = 4m

The number of unavailable books = (5m - 4m) = m

The number of available books unsold = 80

The number of available books sold = (4m - 80)

Now,

$$(4m - 80) - (3m - 120) = 140$$

$$4m - 220 = 3m - 120$$

$$m = 100$$

Now,

The ratio of the total number of books published in March to April = 100:150 = 2:3. Let it be 2n and 3n respectively.

March

@QuizByMickey

The total number of books published = $2n$

The number of available books = 50% of $2n = n$

The number of unavailable books = $(2n - n) = n$

The number of available books unsold = 50

The number of available books sold = $(n - 50)$

April

The total number of books published = $3n$

The number of available books = 60% of $3n = 1.80n$

The number of unavailable books = $(3n - 1.80n) = 1.20n$

The number of available books unsold = 72

The number of available books sold = $(1.80n - 72)$

Now,

$$(1.80n - 72) - (n - 50) = 74$$

$$1.80n - 146 = n - 50$$

$$0.80n = 96$$

$$n = 120$$

January

The total number of books published = $4m = (4 * 100) = 400$

The number of available books = $3m = (3 * 100) = 300$

The number of unavailable books = $m = 100$

The number of available books sold = $(3m - 120) = (300 - 120) = 180$

February

The total number of books published = $5m = (5 * 100) = 500$

The number of available books = $4m = (4 * 100) = 400$

The number of unavailable books = $m = 100$

The number of available books sold = $(4m - 80) = (400 - 80) = 320$

March

The total number of books published = $2n = (2 * 120) = 240$

The number of available books = $n = 120$

The number of unavailable books = $n = 120$

The number of available books sold = $(n - 50) = (120 - 50) = 70$

@QuizByMickey

April

The total number of books published = $3n = (3 * 120) = 360$

The number of available books = $1.80n = (1.80 * 120) = 216$

The number of unavailable books = $1.20n = (1.20 * 120) = 144$

The number of available books sold = $(1.80n - 72) = (216 - 72) = 144$

May

Let the total number of books published be k .

So, the number of available books = 45% of $k = 0.45k$

And the number of unavailable books = $(k - 0.45k) = 0.55k$

The number of available books unsold = 36

So, the number of available books sold = $(0.45k - 36)$

Now,

$$(0.45k - 36) / 0.55k = 5/11$$

$$4.95k - 396 = 2.75k$$

$$2.20k = 396$$

$$k = 180$$

Therefore,

The total number of books published = $k = 180$

The number of available books = $0.45k = (0.45 * 180) = 81$

The number of unavailable books = $0.55k = (0.55 * 180) = 99$

The number of available books sold = $(0.45k - 36) = (81 - 36) = 45$

Month	The total number of books published	The number of available books	The number of unavailable books	The number of available books sold	The number of available books unsold
January	400	300	100	180	120
February	500	400	100	320	80
March	240	120	120	70	50
April	360	216	144	144	72
May	180	81	99	45	36
Total	1680	1117	563	759	358

Answer: E

June

The total number of books published = 200

The number of unavailable books = Z

The number of available books = (200 - Z)

The number of available books unsold = (Z - 24)

The number of available books sold = $\{(200 - Z) - (Z - 24)\} = (224 - 2Z)$

Now,

$$(224 - 2Z) / (Z - 24) = 9/1$$

$$224 - 2Z = 9Z - 216$$

$$11Z = 440$$

$$Z = 40$$

Therefore, the total number of unavailable books in June = Z = 40.

Hence, the correct answer is option E.

19. Questions

Let the total number of books published in January and February be 4m and 5m, respectively.

January

The total number of books published = 4m

The number of available books = 75% of 4m = 3m

The number of unavailable books = (4m - 3m) = m

The number of available books unsold = 120

The number of available books sold = (3m - 120)

February

The total number of books published = 5m

The number of available books = 80% of 5m = 4m

The number of unavailable books = (5m - 4m) = m

The number of available books unsold = 80

The number of available books sold = (4m - 80)

Now,

$$(4m - 80) - (3m - 120) = 140$$

$$4m - 220 = 3m - 120$$

@QuizByMickey

$$m = 100$$

Now,

The ratio of the total number of books published in March to April = $100:150 = 2:3$. Let it be $2n$ and $3n$ respectively.

March

The total number of books published = $2n$

The number of available books = 50% of $2n = n$

The number of unavailable books = $(2n - n) = n$

The number of available books unsold = 50

The number of available books sold = $(n - 50)$

April

The total number of books published = $3n$

The number of available books = 60% of $3n = 1.80n$

The number of unavailable books = $(3n - 1.80n) = 1.20n$

The number of available books unsold = 72

The number of available books sold = $(1.80n - 72)$

Now,

$$(1.80n - 72) - (n - 50) = 74$$

$$1.80n - 146 = n - 50$$

$$0.80n = 96$$

$$n = 120$$

January

The total number of books published = $4m = (4 * 100) = 400$

The number of available books = $3m = (3 * 100) = 300$

The number of unavailable books = $m = 100$

The number of available books sold = $(3m - 120) = (300 - 120) = 180$

February

The total number of books published = $5m = (5 * 100) = 500$

The number of available books = $4m = (4 * 100) = 400$

The number of unavailable books = $m = 100$

The number of available books sold = $(4m - 80) = (400 - 80) = 320$

@QuizByMickey

March

The total number of books published = $2n = (2 * 120) = 240$

The number of available books = $n = 120$

The number of unavailable books = $n = 120$

The number of available books sold = $(n-50) = (120-50) = 70$

April

The total number of books published = $3n = (3 * 120) = 360$

The number of available books = $1.80n = (1.80 * 120) = 216$

The number of unavailable books = $1.20n = (1.20 * 120) = 144$

The number of available books sold = $(1.80n-72) = (216-72) = 144$

May

Let the total number of books published be k .

So, the number of available books = 45% of $k = 0.45k$

And the number of unavailable books = $(k-0.45k) = 0.55k$

The number of available books unsold = 36

So, the number of available books sold = $(0.45k-36)$

Now,

$$(0.45k-36)/0.55k = 5/11$$

$$4.95k - 396 = 2.75k$$

$$2.20k = 396$$

$$k = 180$$

Therefore,

The total number of books published = $k = 180$

The number of available books = $0.45k = (0.45 * 180) = 81$

The number of unavailable books = $0.55k = (0.55 * 180) = 99$

The number of available books sold = $(0.45k-36) = (81-36) = 45$

@QuizByMickey

Month	The total number of books published	The number of available books	The number of unavailable books	The number of available books sold	The number of available books unsold
January	400	300	100	180	120
February	500	400	100	320	80
March	240	120	120	70	50
April	360	216	144	144	72
May	180	81	99	45	36
Total	1680	1117	563	759	358

Answer: E

The total number of available books in April = 216

The total number of available books in May = 81

Let the number of story and poetry books available in April be P and Q, respectively.

So, the number of story books available in May = (P-72)

And, the number of poetry books available in May = 30% of Q = 0.30Q

Now,

$$P + Q = 216 \dots\dots\dots(i)$$

And,

$$(P-72) + 0.30Q = 81$$

By solving equations (i) and (ii), we get,

$$P + 0.30Q = 153$$

$$0.70Q = 63$$

$$Q = 90$$

Now, put the value of Q in equation (i), we get,

$$P + Q = 216$$

$$P + 90 = 216$$

$$P = 126$$

Therefore, the total number of story books published in April and May together = {P+(P-72)} = {126+(126-72)} = (252-72) = 180.

Hence, the correct answer is option E.

@QuizByMickey

20. Questions

Let the total number of books published in January and February be $4m$ and $5m$, respectively.

January

The total number of books published = $4m$

The number of available books = 75% of $4m = 3m$

The number of unavailable books = $(4m - 3m) = m$

The number of available books unsold = 120

The number of available books sold = $(3m - 120)$

February

The total number of books published = $5m$

The number of available books = 80% of $5m = 4m$

The number of unavailable books = $(5m - 4m) = m$

The number of available books unsold = 80

The number of available books sold = $(4m - 80)$

Now,

$$(4m - 80) - (3m - 120) = 140$$

$$4m - 220 = 3m - 120$$

$$m = 100$$

Now,

The ratio of the total number of books published in March to April = $100:150 = 2:3$. Let it be $2n$ and $3n$ respectively.

March

The total number of books published = $2n$

The number of available books = 50% of $2n = n$

The number of unavailable books = $(2n - n) = n$

The number of available books unsold = 50

The number of available books sold = $(n - 50)$

April

The total number of books published = $3n$

The number of available books = 60% of $3n = 1.80n$

The number of unavailable books = $(3n - 1.80n) = 1.20n$

@QuizByMickey

The number of available books unsold = 72

The number of available books sold = $(1.80n-72)$

Now,

$$(1.80n-72) - (n-50) = 74$$

$$1.80n - 146 = n - 50$$

$$0.80n = 96$$

$$n = 120$$

January

The total number of books published = $4m = (4 * 100) = 400$

The number of available books = $3m = (3 * 100) = 300$

The number of unavailable books = $m = 100$

The number of available books sold = $(3m-120) = (300-120) = 180$

February

The total number of books published = $5m = (5 * 100) = 500$

The number of available books = $4m = (4 * 100) = 400$

The number of unavailable books = $m = 100$

The number of available books sold = $(4m-80) = (400-80) = 320$

March

The total number of books published = $2n = (2 * 120) = 240$

The number of available books = $n = 120$

The number of unavailable books = $n = 120$

The number of available books sold = $(n-50) = (120-50) = 70$

April

The total number of books published = $3n = (3 * 120) = 360$

The number of available books = $1.80n = (1.80 * 120) = 216$

The number of unavailable books = $1.20n = (1.20 * 120) = 144$

The number of available books sold = $(1.80n-72) = (216-72) = 144$

May

Let the total number of books published be k .

So, the number of available books = 45% of $k = 0.45k$

And the number of unavailable books = $(k-0.45k) = 0.55k$

@QuizByMickey

The number of available books unsold = 36

So, the number of available books sold = $(0.45k - 36)$

Now,

$$(0.45k - 36) / 0.55k = 5/11$$

$$4.95k - 396 = 2.75k$$

$$2.20k = 396$$

$$k = 180$$

Therefore,

The total number of books published = $k = 180$

The number of available books = $0.45k = (0.45 * 180) = 81$

The number of unavailable books = $0.55k = (0.55 * 180) = 99$

The number of available books sold = $(0.45k - 36) = (81 - 36) = 45$

Month	The total number of books published	The number of available books	The number of unavailable books	The number of available books sold	The number of available books unsold
January	400	300	100	180	120
February	500	400	100	320	80
March	240	120	120	70	50
April	360	216	144	144	72
May	180	81	99	45	36
Total	1680	1117	563	759	358

Answer: A

July

The total number of available books (sold + unsold) = 100

Let the number of available books sold to females be R.

So, the number of available books sold to males = $(R + 40)$

Therefore, the total number of available books sold = $\{R + (R + 40)\} = (2R + 40)$

The number of available books unsold = R

Now,

$$(2R + 40) + R = 100$$

$$3R = 60$$

@QuizByMickey

$$R = 20$$

Therefore, the number of available books sold to males = $(R+40) = (20+40) = 60$.

Hence, the correct answer is option A.

@QuizByMickey